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Smith, Jr. et al.

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(45) **Date of Patent: Aug. 26, 2025**

(54) **CIRCUITS AND GROUP III-NITRIDE TRANSISTORS WITH BURIED P-LAYERS AND CONTROLLED GATE VOLTAGES AND METHODS THEREOF**

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(73) Assignee: **Wolfspeed, Inc.**, Durham, NC (US)

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(65) **Prior Publication Data**

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(51) **Int. Cl.**

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H03K 17/687 (2006.01)
H10D 30/01 (2025.01)
H10D 62/824 (2025.01)
H10D 62/85 (2025.01)
H10D 64/00 (2025.01)

(52) **U.S. Cl.**

CPC ***H10D 30/475*** (2025.01); ***H03K 17/687*** (2013.01); ***H10D 30/015*** (2025.01); ***H10D 62/824*** (2025.01); ***H10D 62/8503*** (2025.01); ***H10D 64/111*** (2025.01)

(58) **Field of Classification Search**

CPC H01L 29/7786; H01L 29/2003; H01L 29/205; H01L 29/402; H01L 29/66462

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,289,015	A	2/1994	Chirovsky et al.
5,318,915	A	6/1994	Baliga et al.
6,639,277	B2	10/2003	Rumennik et al.
6,956,239	B2	10/2005	Sriram
7,646,043	B2	1/2010	Sriram et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1717811	A	1/2006
CN	102623493	A	8/2012

(Continued)

OTHER PUBLICATIONS

International Patent Application No. PCT/US2022/028963; Int'l Search Report and the Written Opinion; dated Sep. 1, 2022; 12 pages.

(Continued)

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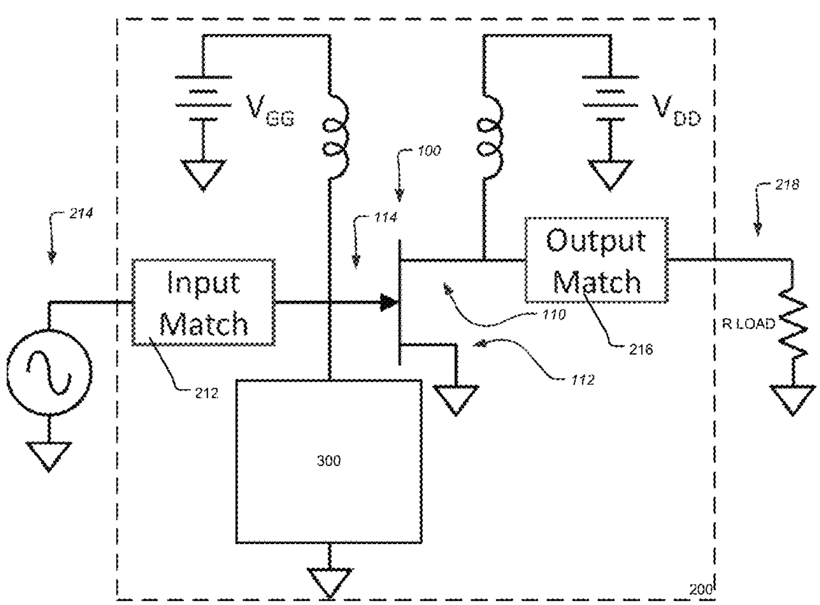
(57) **ABSTRACT**

An apparatus for reducing lag includes a substrate; a group III-Nitride barrier layer; a source electrically coupled to the group III-Nitride barrier layer; a gate on the group III-Nitride barrier layer; a drain electrically coupled to the group III-Nitride barrier layer; a p-region being arranged at or below the group III-Nitride barrier layer; and a gate control circuit configured to control a gate voltage of the gate. Additionally, at least a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain.

36 Claims, 17 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

7,696,566	B2	4/2010	Disney	
7,812,369	B2	10/2010	Chini et al.	
8,698,162	B2	4/2014	Lee et al.	
8,759,879	B1	6/2014	Tipirneni et al.	
8,890,211	B1	11/2014	Mayo et al.	
8,901,609	B1	12/2014	Chen et al.	
9,337,332	B2	5/2016	Chu et al.	
9,484,862	B2	11/2016	De Rooij et al.	
9,621,153	B2	4/2017	Ikeda et al.	
10,290,730	B1	5/2019	Yang et al.	
2002/0066960	A1	6/2002	Ring	
2003/0141518	A1	7/2003	Yokogawa et al.	
2004/0099888	A1	5/2004	Sriram	
2004/0124435	A1	7/2004	D'Evelyn et al.	
2004/0149993	A1	8/2004	McClure et al.	
2004/0201037	A1	10/2004	Fareed et al.	
2005/0121694	A1	6/2005	Mun et al.	
2005/0224809	A1	10/2005	Sriram	
2006/0125001	A1	6/2006	Sriram	
2007/0051977	A1	3/2007	Saito et al.	
2007/0059873	A1	3/2007	Chini et al.	
2007/0246713	A1	10/2007	Arnold et al.	
2007/0292999	A1	12/2007	Henning et al.	
2008/0315257	A1	12/2008	Shiraishi	
2009/0120924	A1	5/2009	Moffatt et al.	
2010/0084742	A1	4/2010	Sazawa et al.	
2011/0215338	A1	9/2011	Zhang	
2011/0260174	A1	10/2011	Hebert	
2012/0025206	A1	2/2012	Nakata et al.	
2012/0068227	A1	3/2012	Hikita et al.	
2012/0153300	A1	6/2012	Lidow et al.	
2012/0187451	A1*	7/2012	Saito	H10D 64/254 257/192
2012/0217511	A1	8/2012	Renaud et al.	
2012/0286355	A1*	11/2012	Mauder	H10D 12/481 257/330
2012/0319127	A1	12/2012	Chowdhury et al.	
2013/0049014	A1	2/2013	Aigo et al.	
2013/0062671	A1	3/2013	Saito et al.	
2013/0153967	A1*	6/2013	Curatola	H01L 29/66462 257/E29.246
2013/0221320	A1	8/2013	Li et al.	
2014/0117381	A1	5/2014	Kang et al.	
2014/0145203	A1	5/2014	Escoffier et al.	
2014/0253241	A1	9/2014	Lee et al.	
2014/0264379	A1	9/2014	Kub et al.	
2014/0264960	A1	9/2014	Ring et al.	
2015/0028345	A1	1/2015	Wong et al.	
2015/0028350	A1	1/2015	Suvorov et al.	
2015/0115325	A1	4/2015	Vielemeyer	
2015/0145032	A1	5/2015	Quay et al.	
2015/0236146	A1	8/2015	Liu et al.	
2015/0318387	A1	11/2015	Chiu et al.	
2015/0349117	A1	12/2015	Chu et al.	
2015/0372124	A1	12/2015	Isobe et al.	
2016/0035870	A1	2/2016	Wu et al.	
2016/0071967	A1	3/2016	Prechtl et al.	
2016/0086878	A1	3/2016	Otremba et al.	
2016/0254377	A1	9/2016	Morancho et al.	
2016/0293709	A1	10/2016	Nakayama et al.	
2016/0351567	A1	12/2016	Schmid	
2017/0018638	A1	1/2017	Teo et al.	
2017/0033210	A1	2/2017	Curatola et al.	
2017/0194471	A1	7/2017	Pei	
2017/0244407	A1	8/2017	Prechtl et al.	
2017/0250273	A1	8/2017	Schultz et al.	
2017/0365702	A1	12/2017	Prechtl et al.	
2017/0373176	A1	12/2017	Sriram	
2017/0373178	A1	12/2017	Sriram et al.	
2018/0083105	A1	3/2018	Nagy et al.	
2018/0167043	A1*	6/2018	Van Zyl	H03F 3/2176
2018/0204940	A1	7/2018	Schultz et al.	
2018/0269282	A1	9/2018	Yang et al.	
2019/0229187	A1	7/2019	Sriram et al.	
2019/0237569	A1	8/2019	Sriram et al.	
2019/0356274	A1	11/2019	Zhu et al.	
2020/0219987	A1	7/2020	Lee et al.	
2020/0357907	A1	11/2020	Udrea et al.	
2021/0126120	A1	4/2021	Piedra et al.	
2021/0167199	A1	6/2021	Sriram et al.	

FOREIGN PATENT DOCUMENTS

CN	103178108	A	6/2013
CN	103329256	A	9/2013
CN	105405877	A	3/2016
CN	109564940	A	4/2019
CN	212062440	U	12/2020
EP	3276670	A1	1/2018
JP	2002-076024	A	3/2002
JP	2006-507683	A	3/2006
JP	2007-103451	A	4/2007
JP	2008-112868	A	5/2008
JP	2011-151176	A	8/2011
JP	2012-156320	A	8/2012
JP	2012-231002	A	11/2012
JP	2012-238751	A	12/2012
JP	2012-248632	A	12/2012
JP	2012-248753	A	12/2012
JP	2013-038250	A	2/2013
JP	2013-106022	A	5/2013
JP	2013-131755	A	7/2013
JP	2013-229449	A	11/2013
JP	2014-504013	A	2/2014
JP	2014-520405	A	8/2014
JP	2016-537828	A	12/2016
JP	2017-059786	A	3/2017
JP	2017-528912	A	9/2017
JP	2019-091757	A	6/2019
JP	2019-519121	A	7/2019
KR	2015-0120358	A	10/2015
WO	WO 2014/125094	A1	8/2014
WO	WO 2015/013620	A1	1/2015
WO	WO 2017/223403	A1	12/2017
WO	WO 2020/159934	A1	8/2020

OTHER PUBLICATIONS

European Patent Application No. 20749178.8; Partial Search Report; dated Sep. 28, 2022; 10 pages.

Chiu et al.; "Analysis of the Back-Gate Effect in Normally OFF p-GaN Gate High-Electron Mobility Transistor"; IEEE Transactions on Electron Devices; vol. 62 No. 2; Feb. 2015; p. 507-511.

R R. Pela, C. Caetano, M. Marques, L. G. Ferreira, J. Furthmuller, and L. K. Teles, "Accurate band gaps of AlGaN, InGaN, and AlInN alloys calculations based on LDA-1/2 approach", Appl. Phys. Lett. 98, 151907 (2011) Appl. Phys. Lett. 98, 151907; 3 pages. (Year: 2011).

Shailaja P. Rao, "Implant Annealing of al dopants in silicon carbide using silane overpressure", Scholar Commons, University of South Florida, 2005, pp. 21-22; Retrieved from http://scholarscommons.usf.edu/cgi/viewcontent.cgiarticle=1828 context=e-td on Aug. 28, 2017 [Cited in related U.S. Appl. No. 15/192,545 and U.S. Appl. No. 15/192,545].

Wendt et al., "FIB Milling and Canneling" Nov. 1, 2008, Figure 1, p. 1; Retrieved from http://www.imaging-git.com/science/scanning-probe-microscopy/fib-milling-- - and-canneling on Aug. 28, 2017 [Cited in related U.S. Appl. No. 15/192,545 and U.S. Appl. No. 15/192,545].

Danielsson et al., "Fabrication and Characterization of Heterojunction Diodes with HVPE-Grown GaN on 4H—SiC" IEEE Transactions on Electron Devices, vol. 48, No. 3, Mar. 2001, pp. 444-449 [Cited in related U.S. Appl. No. 15/192,545 and U.S. Appl. No. 15/192,545].

Pal et al. "High electron Mobility transistor (HEMT)" IJSRET, vol. 1, Issue. 1, Mar. 2012, pp. 043-046 (Year: 2012) [Cited in related U.S. Appl. No. 15/192,545].

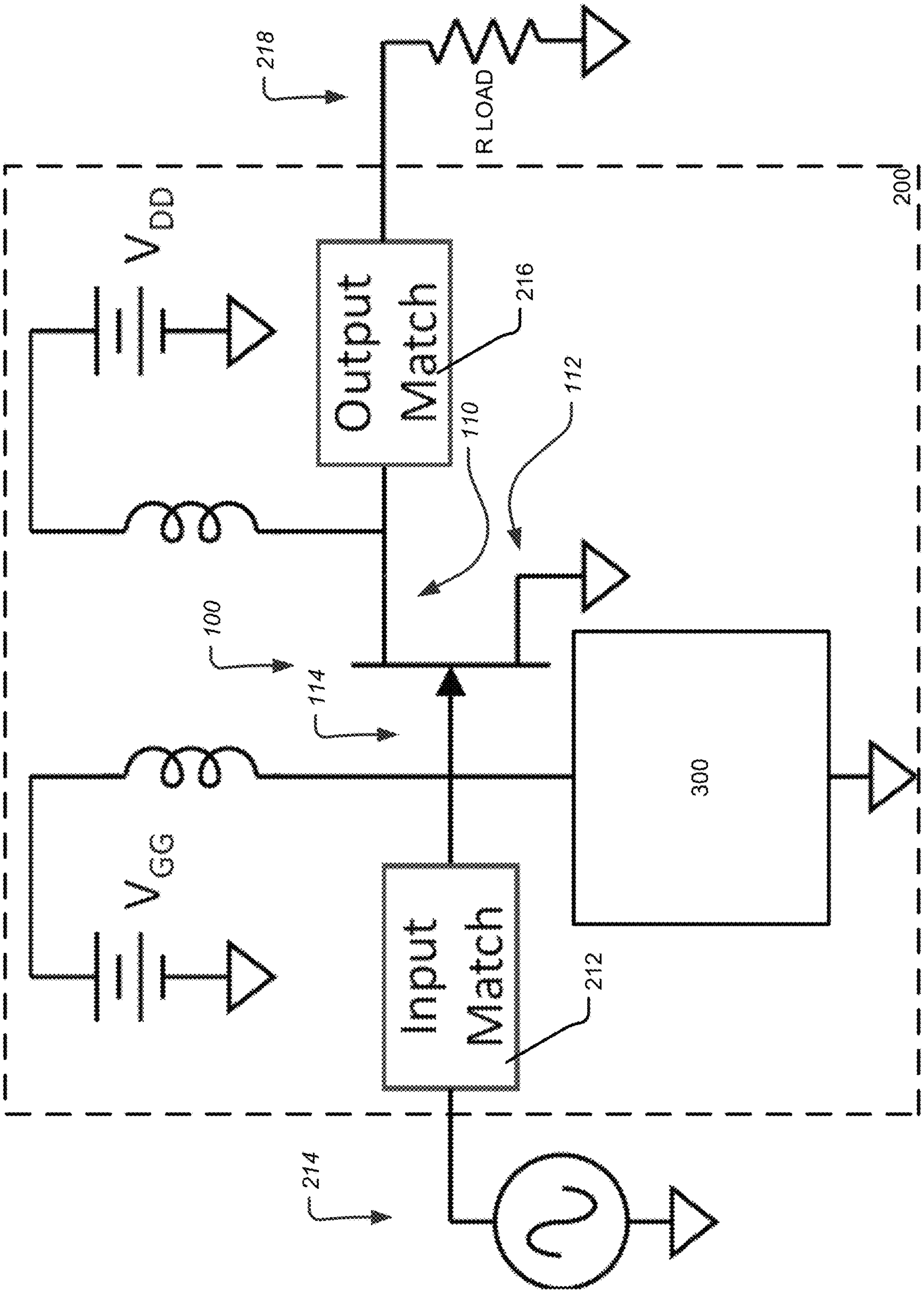
International Search Report and Written Opinion dated Sep. 21, 2007 for PCT/US2017/38937, filed Jun. 23, 2017.; 13 pages.

(56) **References Cited**

OTHER PUBLICATIONS

Final Office Action issued in U.S. Appl. No. 15/192,545, dated Apr. 12, 2018; 21 pages.
Kayali et al., "GaAs MMI C Reliability Assurance Guideline forSpace Applications" JPL Publication 96-25, Nat'l Aeronautics and Space Administration, Dec. 15, 1996, 221 pages.
Sun et al.; "Impact of Traps on the Adjacent Channel Power Ratios of GaN HEMTs"; IEEE Electron Device Letters; vol. 41 No. 6; Jun. 2020; p. 816-819.
European Patent Application No. 20749178.8; Extended Search Report; dated Feb. 2, 2023; 11 pages.
International Patent Application No. PCT/US2022/13085; Int'l Search Report and the Written Opinion; dated Jun. 28, 2022; 21 pages.
International Patent Application No. PCT/US2022/18489; Int'l Search Report and the Written Opinion; dated May 23, 2022; 22 pages.
International Search Report and Written Opinion issued in PCT/US2020/015331, dated Mar. 4, 2020; 8 pages.
European Patent Application No. 22753113.4; Extended Search Report; dated Dec. 3, 2024; 9 pages.
European Patent Application No. 22805124.9; Extended Search Report; dated Feb. 21, 2025; 10 pages.
European Patent Application No. 22805206.4; Extended Search Report; dated Feb. 21, 2025; 10 pages.

* cited by examiner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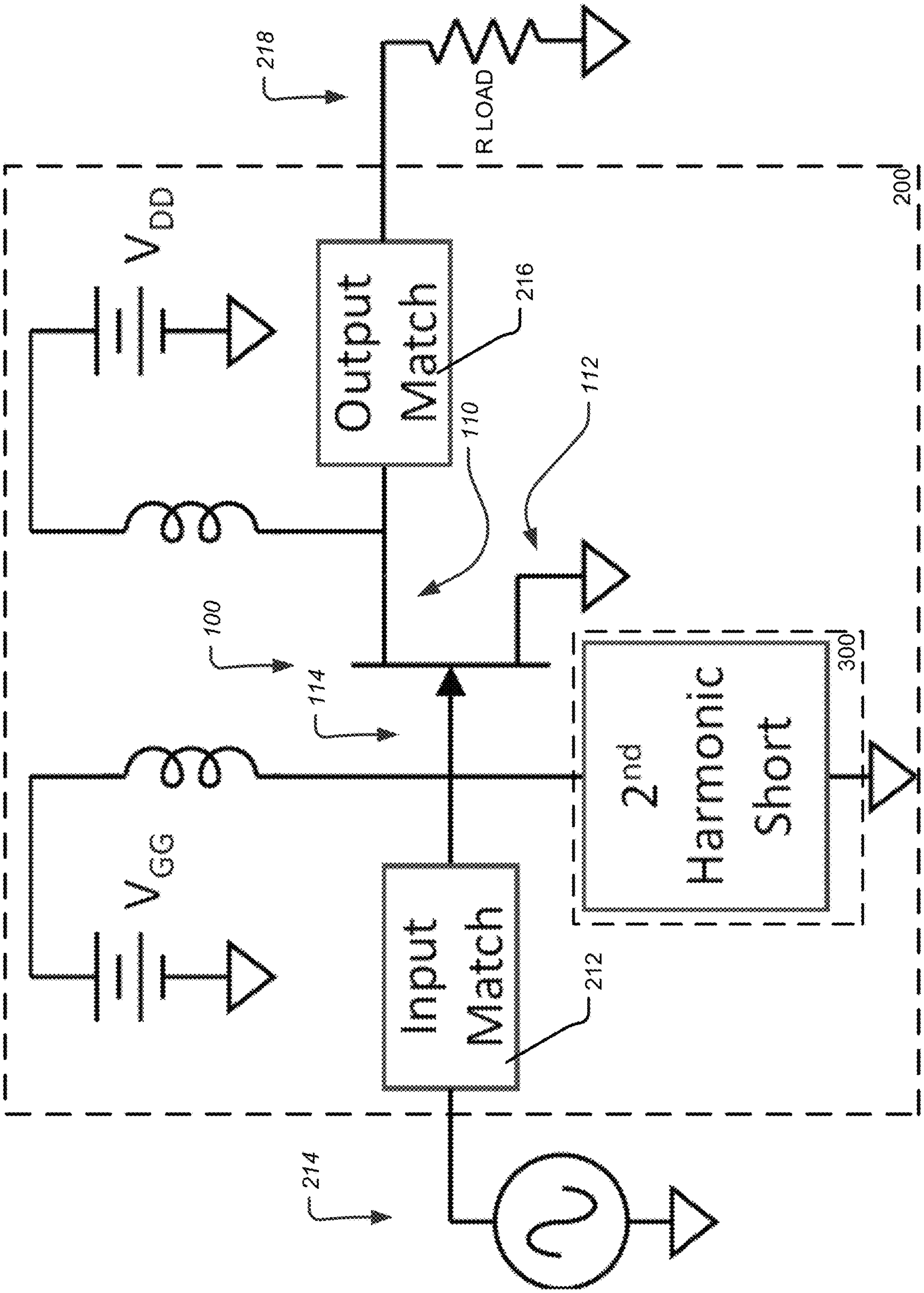
216
112
110
100
114
300
V_{GG}
Input Match
212
214

FIG. 1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114
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2nd Harmonic Short
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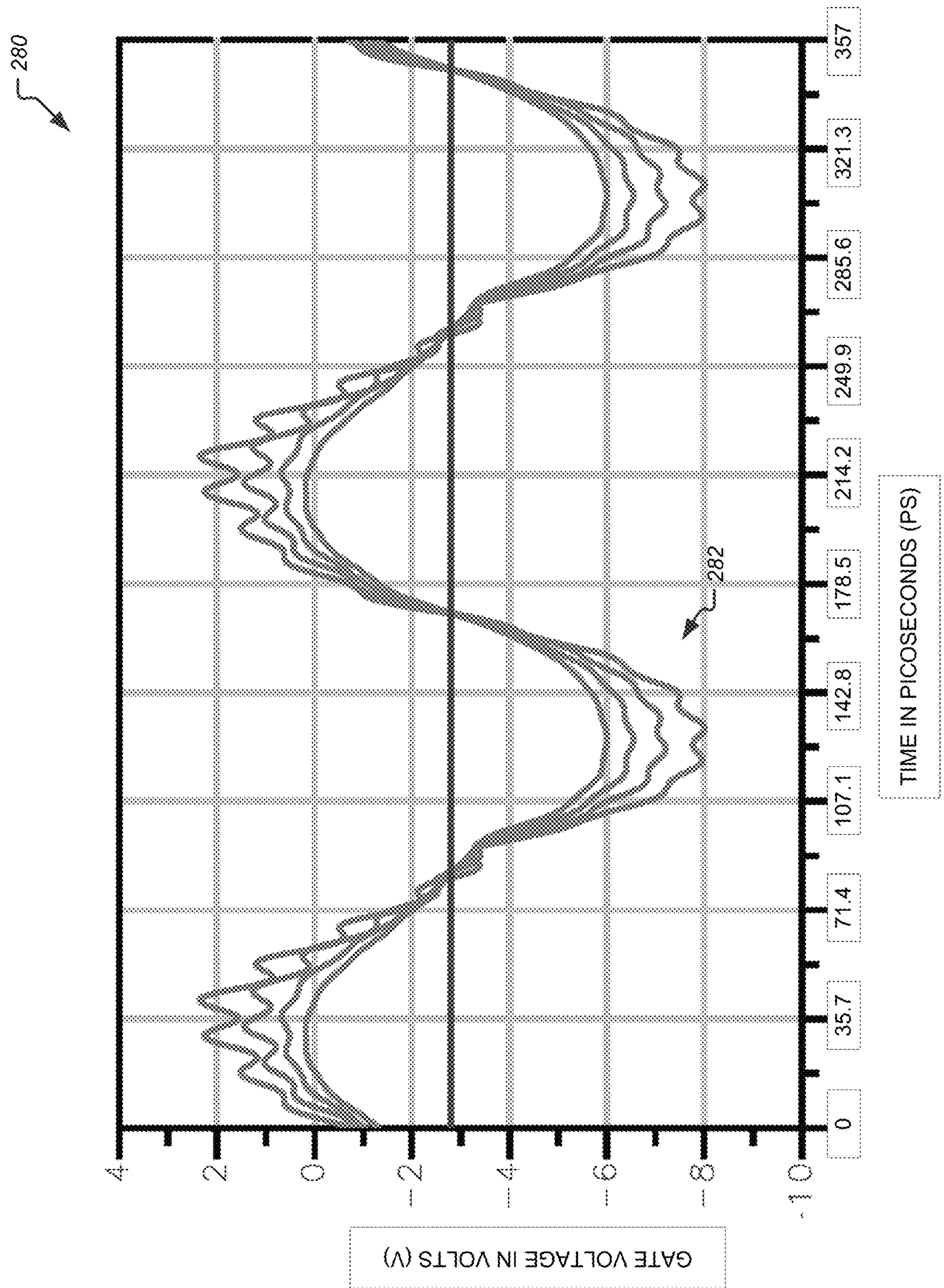
Input Match
212
214

FIG. 2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282
GATE VOLTAGE IN VOLTS (V)
4
2
0
−2
−4
−6
−8
-10
TIME IN PICOSECONDS (PS)
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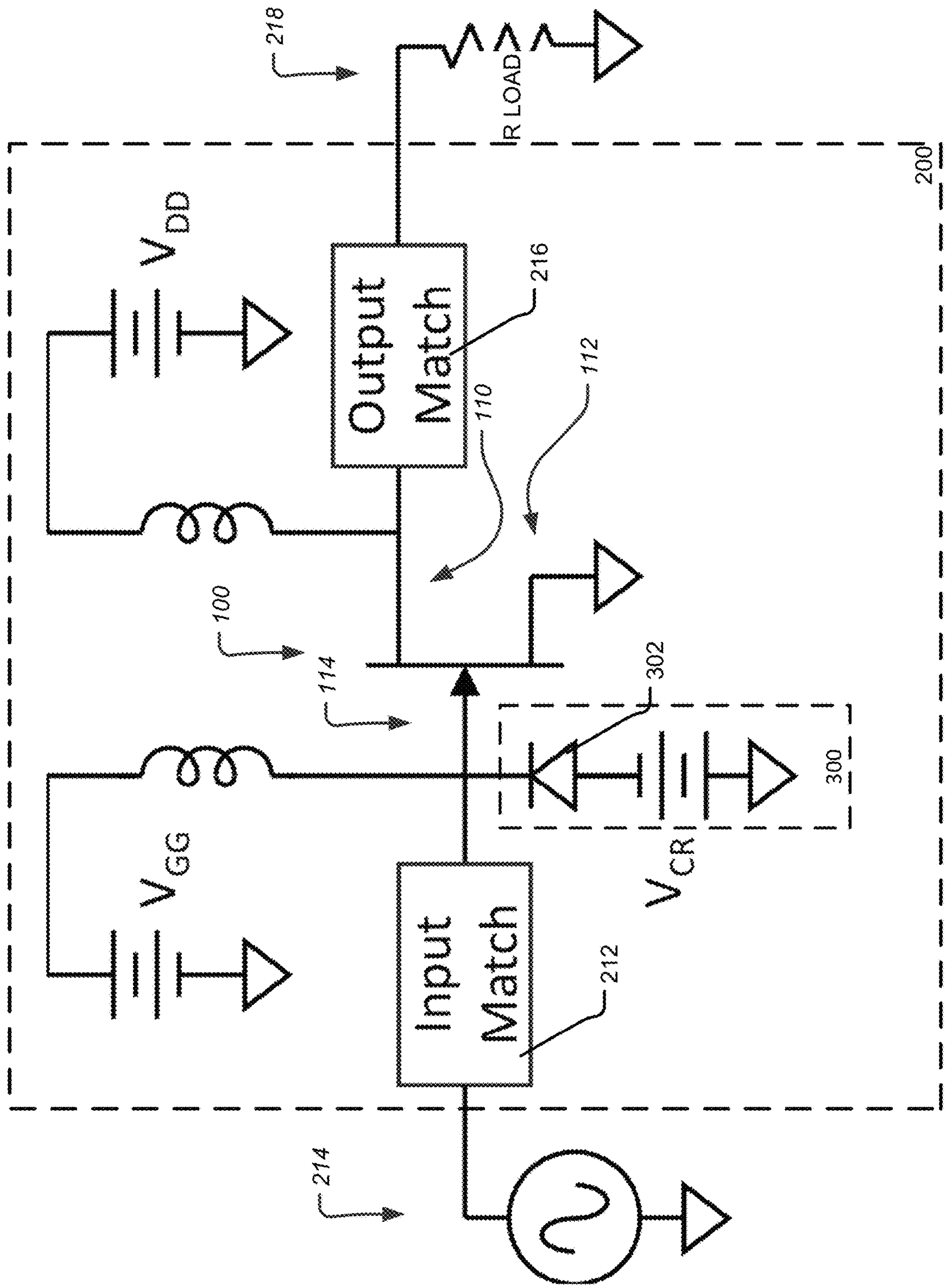
35.7
71.4
107.1
142.8
178.5
214.2
249.9
285.6
321.3
357

FIG. 3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R LOAD
200
V_{DD}
Output Match
216
110
112
100
114
302
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V_{GG}
V_{CR}
Input Match
212
214

FIG. 4

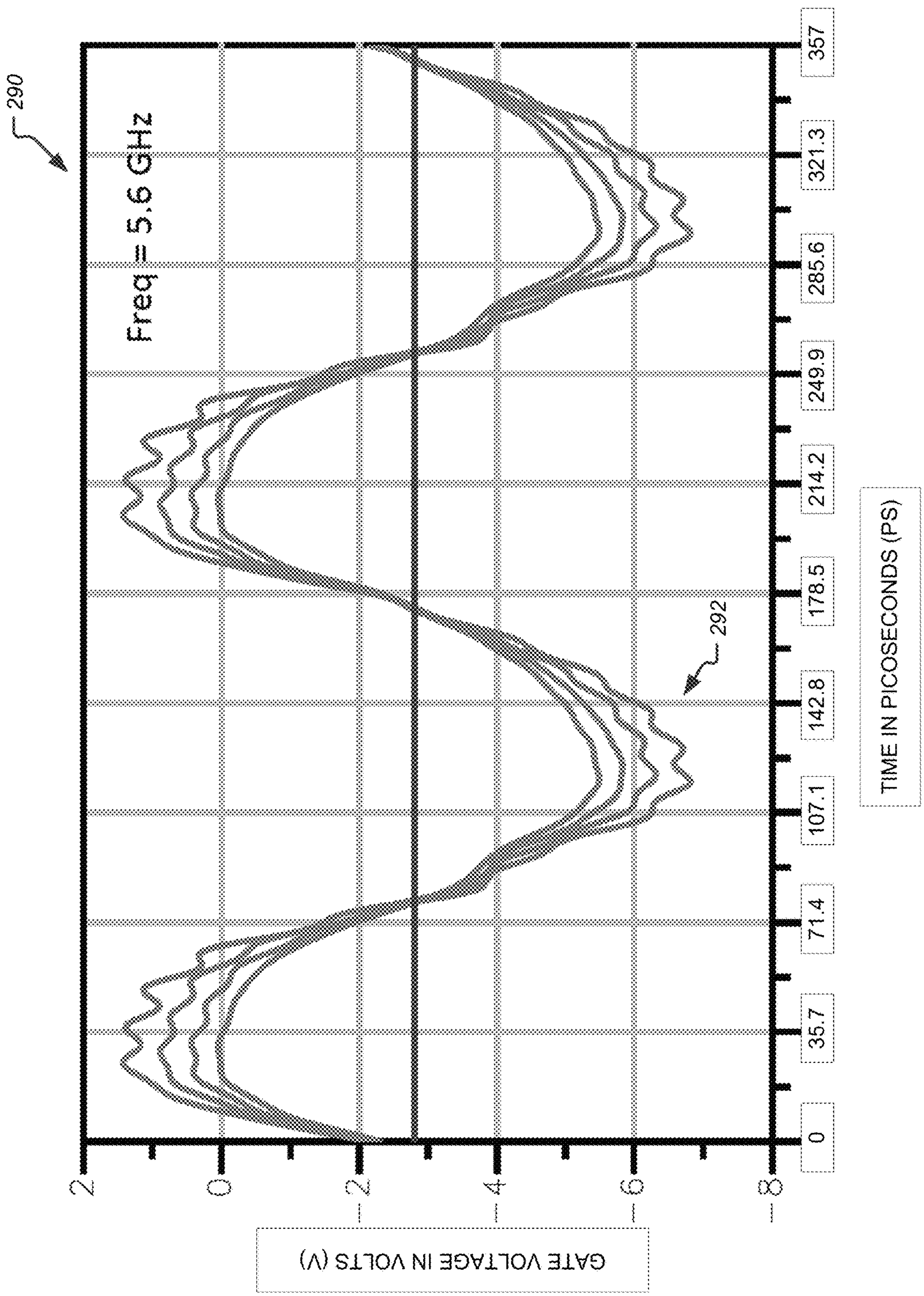

290
Freq = 5.6 GHz
292
GATE VOLTAGE IN VOLTS (V)
2
0
−2
−4
−6
−8
TIME IN PICOSECONDS (PS)
0
35.7
71.4
107.1
142.8
178.5
214.2
249.9
285.6
321.3
357

FIG. 5

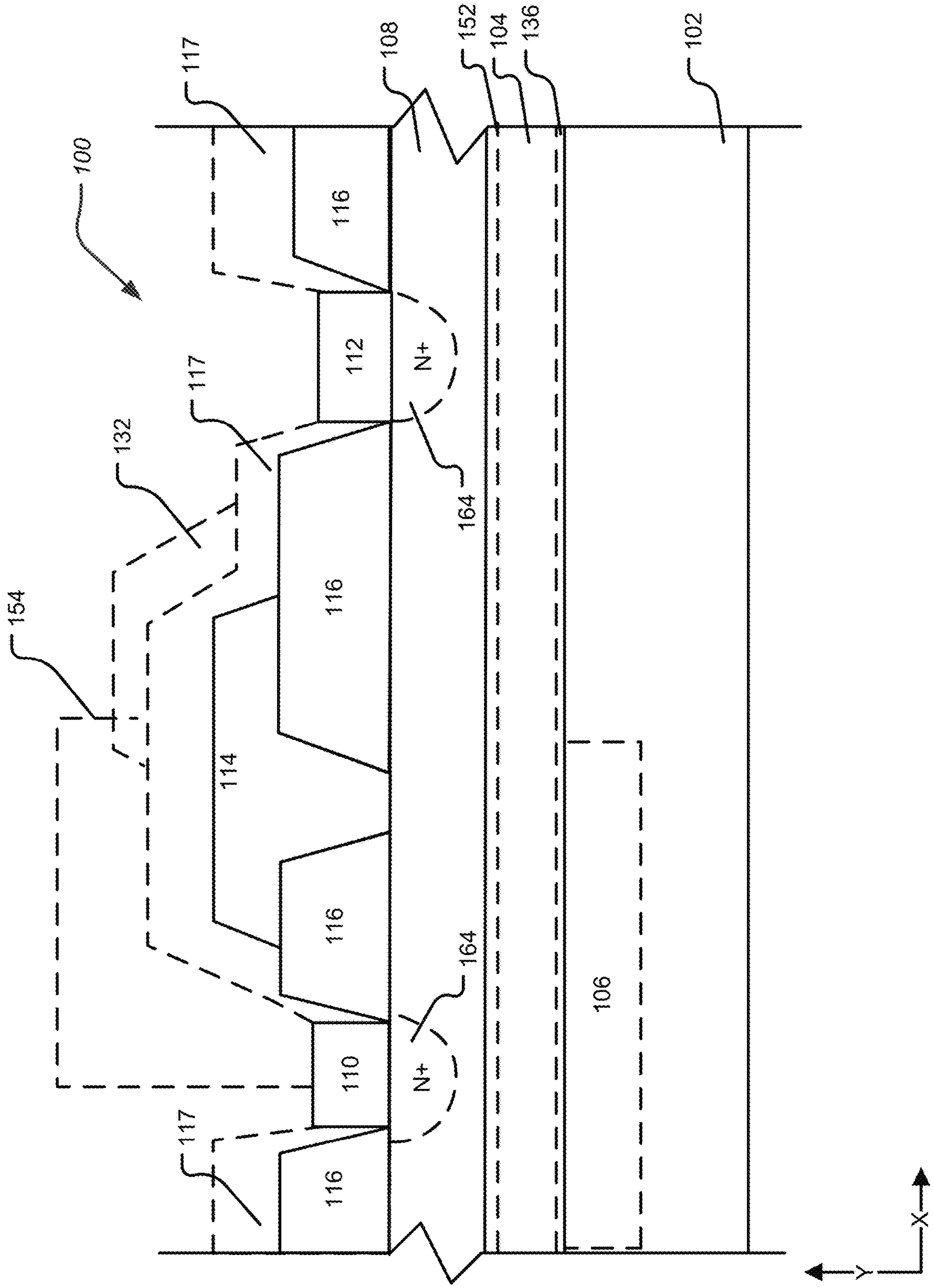
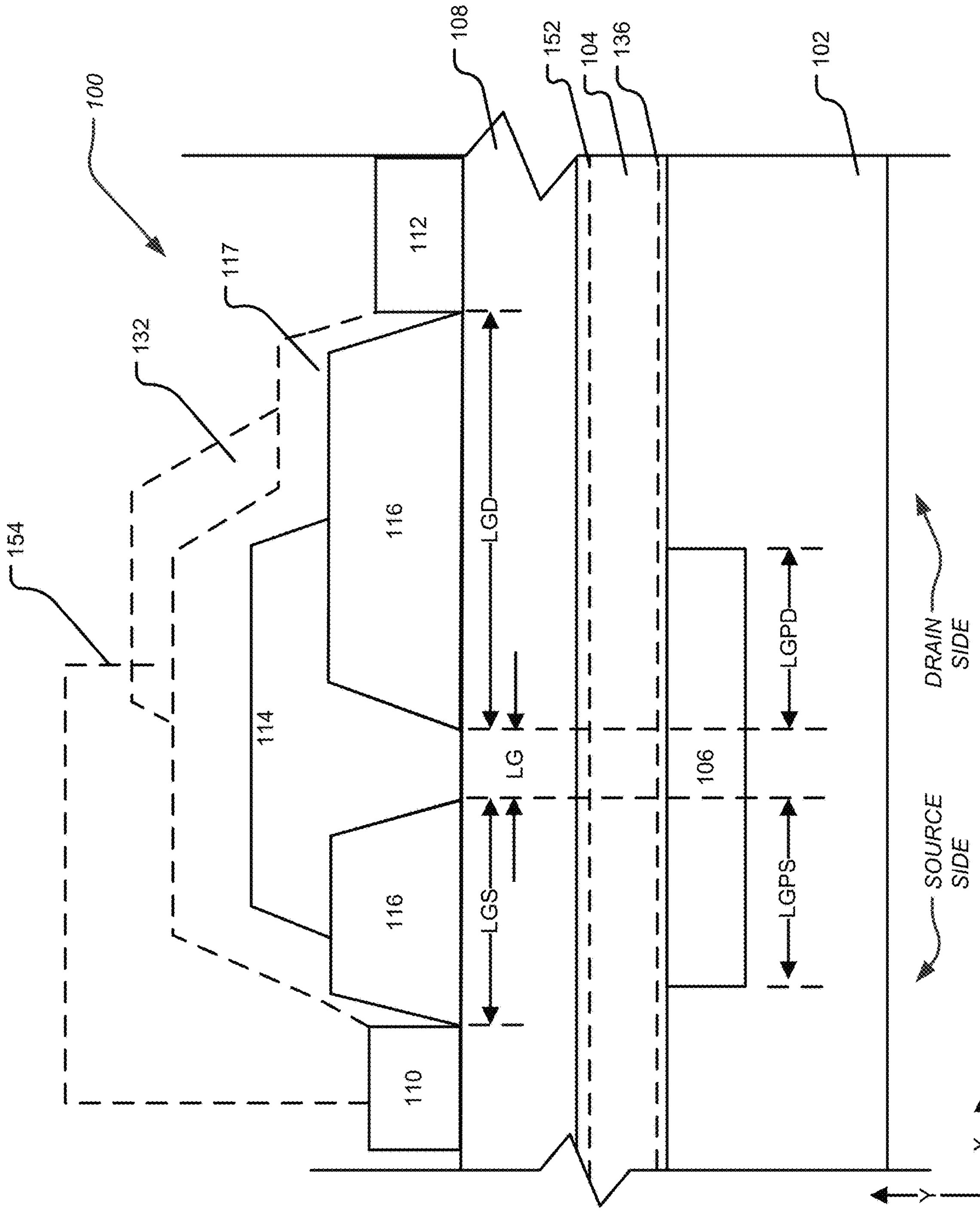


FIG. 6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108
152
104
136
102
112
117
132
154
116
114
116
110
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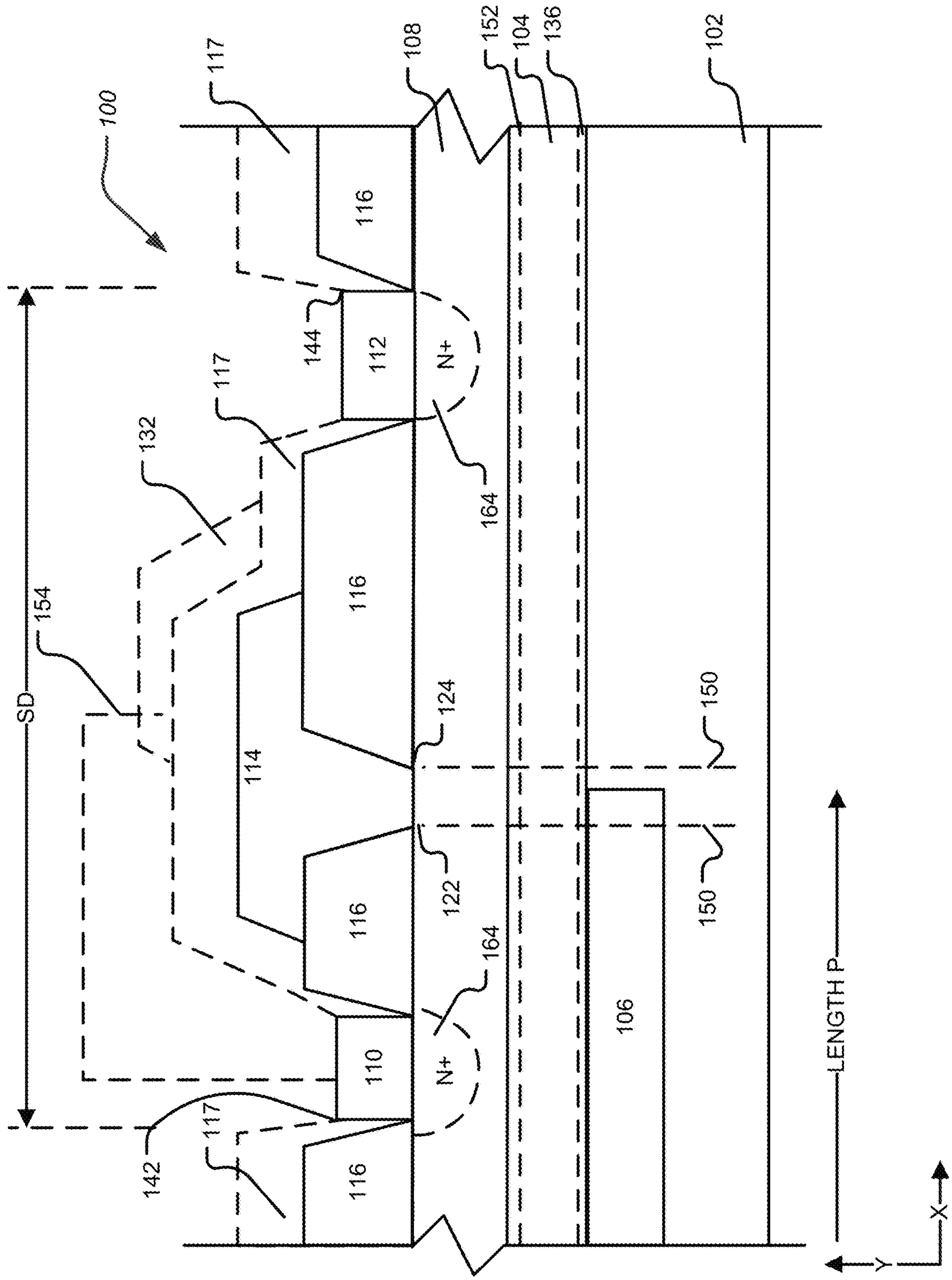
LG
LGS
LGPD
LGPS
106
DRAIN SIDE
SOURCE SIDE
X
Y

FIG. 7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112
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117
132
164
116
154
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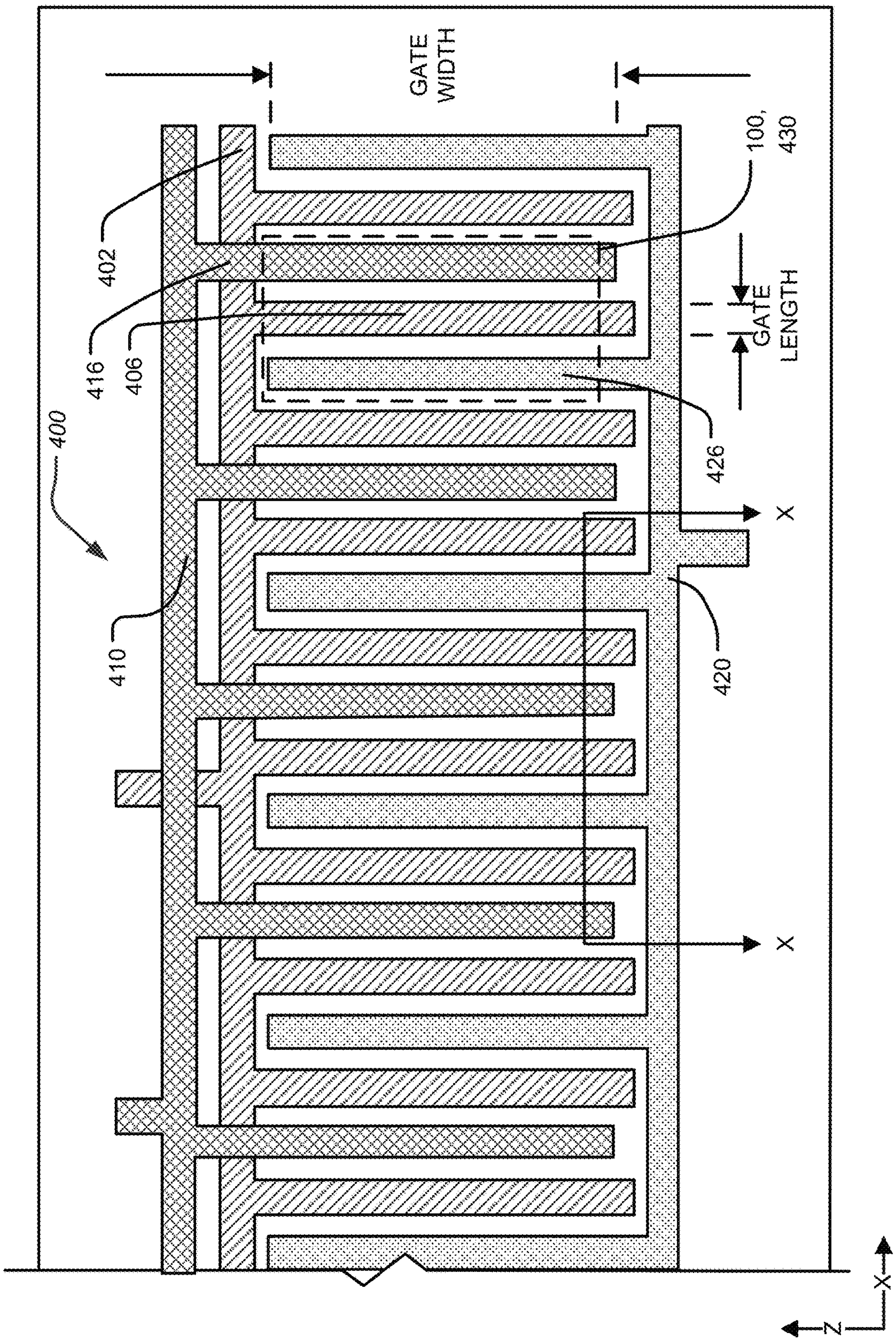
114
122
116
164
150
N+
110
106
142
117
116
LENGTH P
X
Y

FIG. 8


GATE WIDTH
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GATE LENGTH
402
416
406
400
426
X
410
420
X
X
Z

FIG. 9

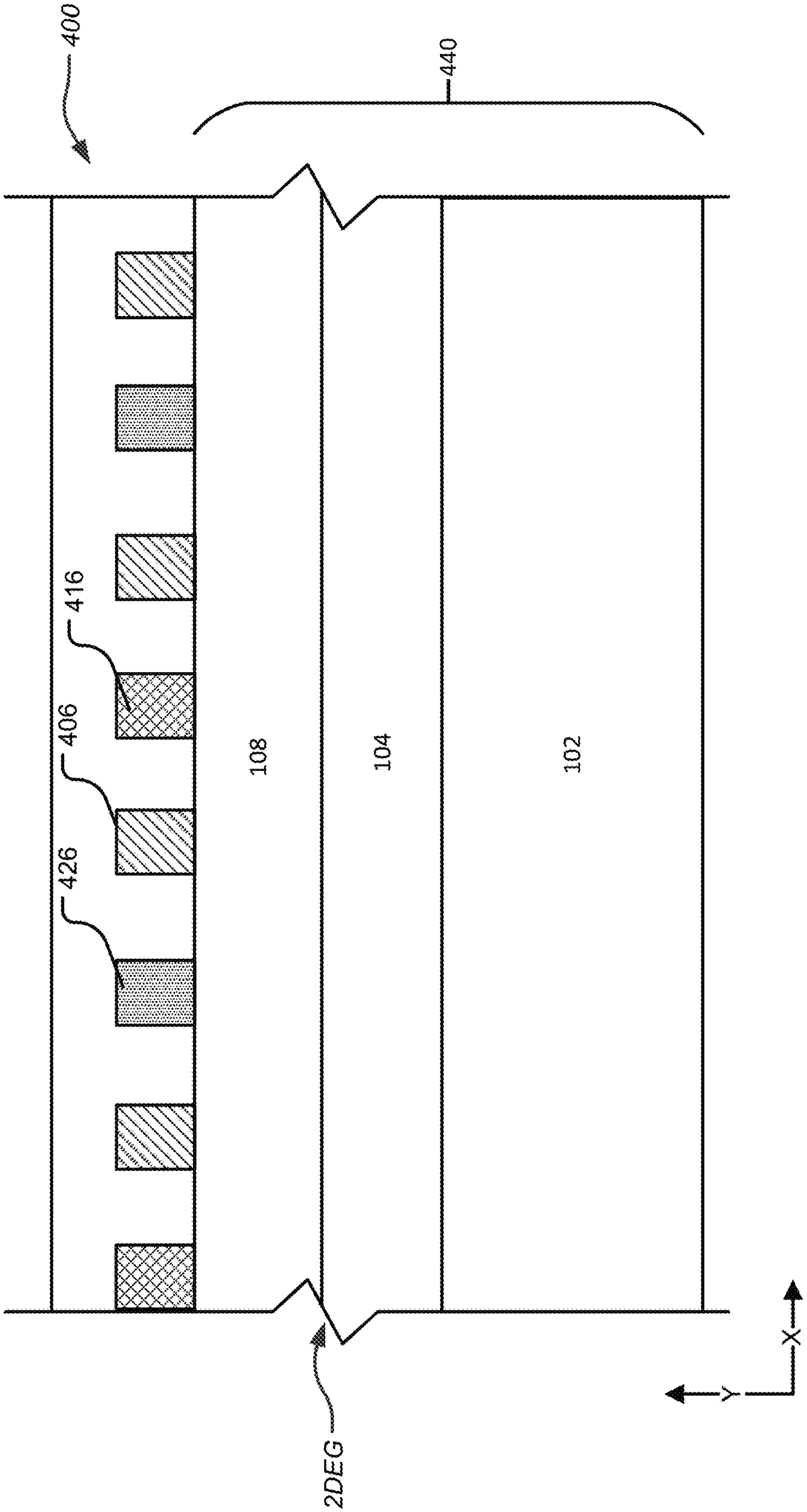
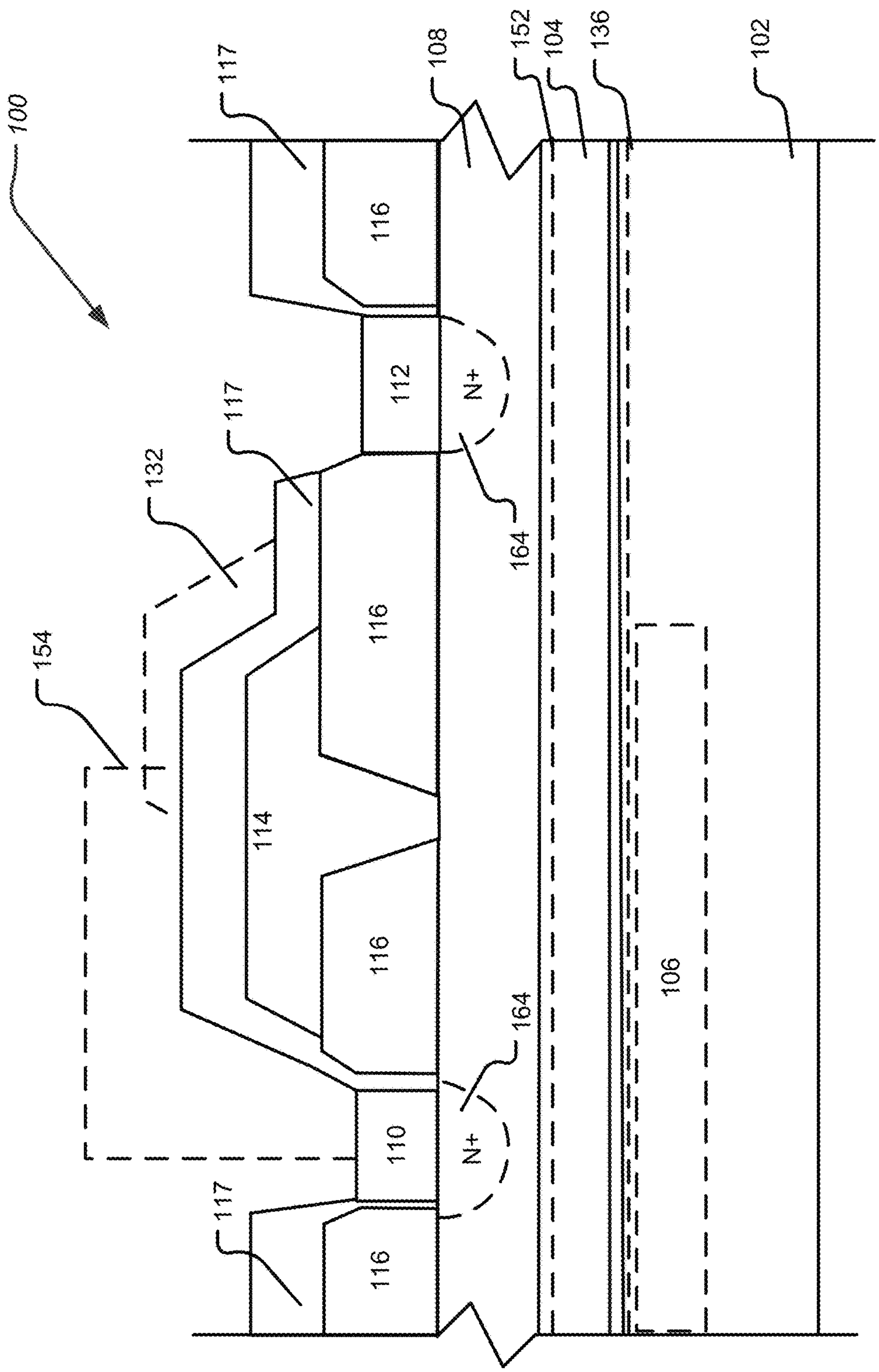


FIG. 10


100
117
116
108
152
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136
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112
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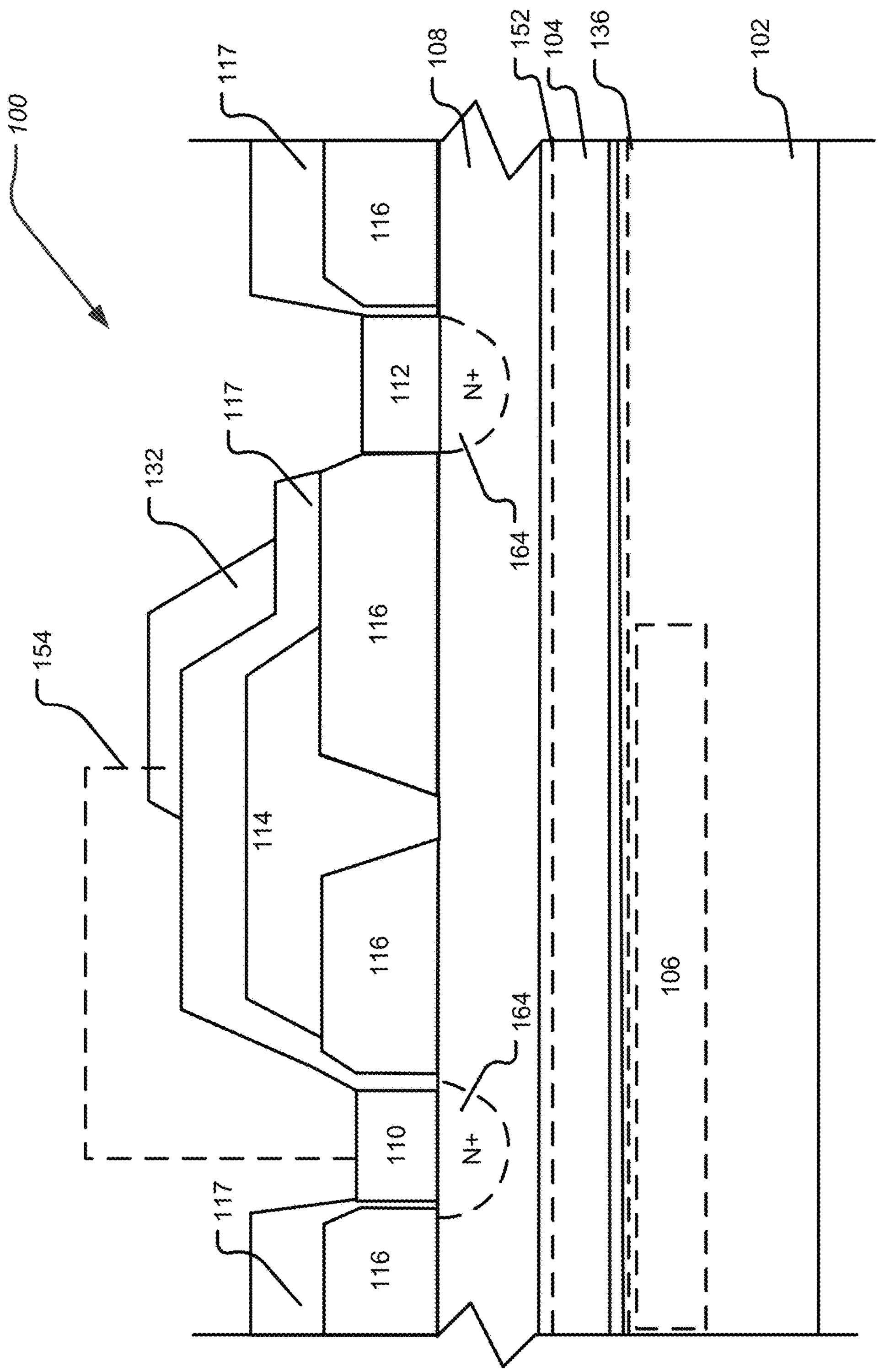
117
132
116
154
114
116
164
110
N+
117
116
106

FIG. 11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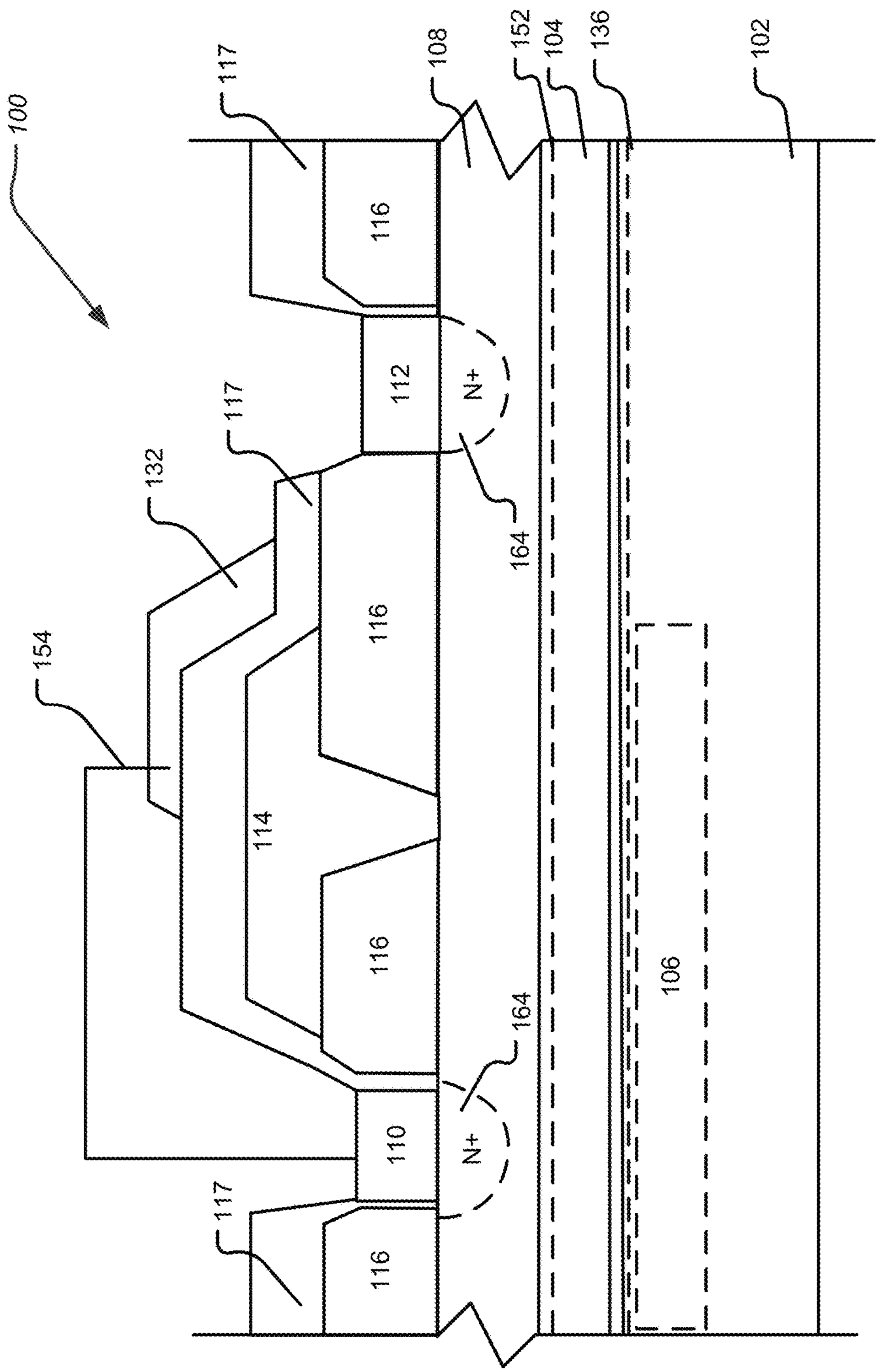
116
108
152
104
136
102
112
N+
117
132
164
116
154
114
116
106
164
N+
110
117
116

FIG. 12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136
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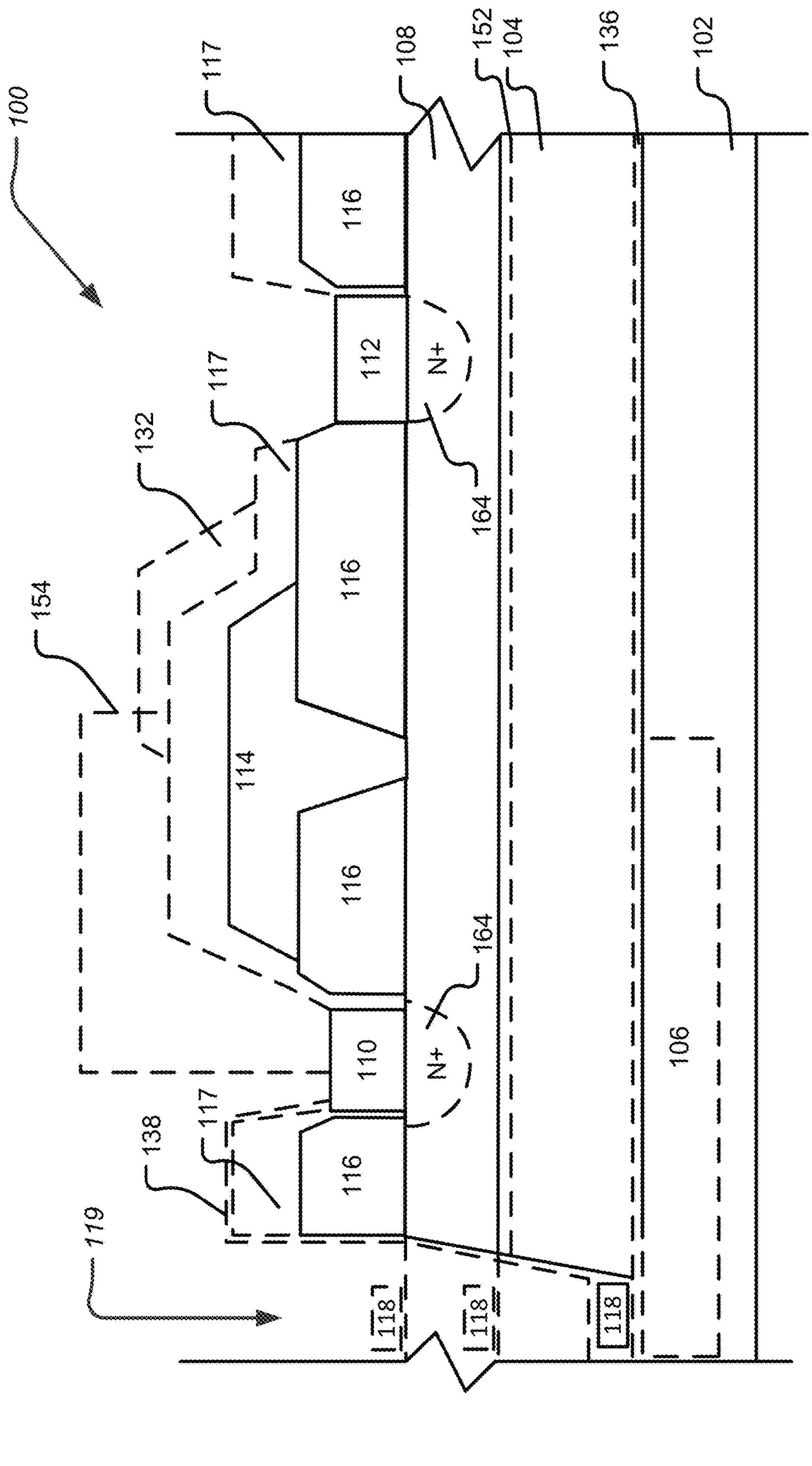
114
116
106
164
110
N+
117
116

FIG. 13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152
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N+
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132
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N+
106
138
117
116
119
118
118
118

FIG. 14

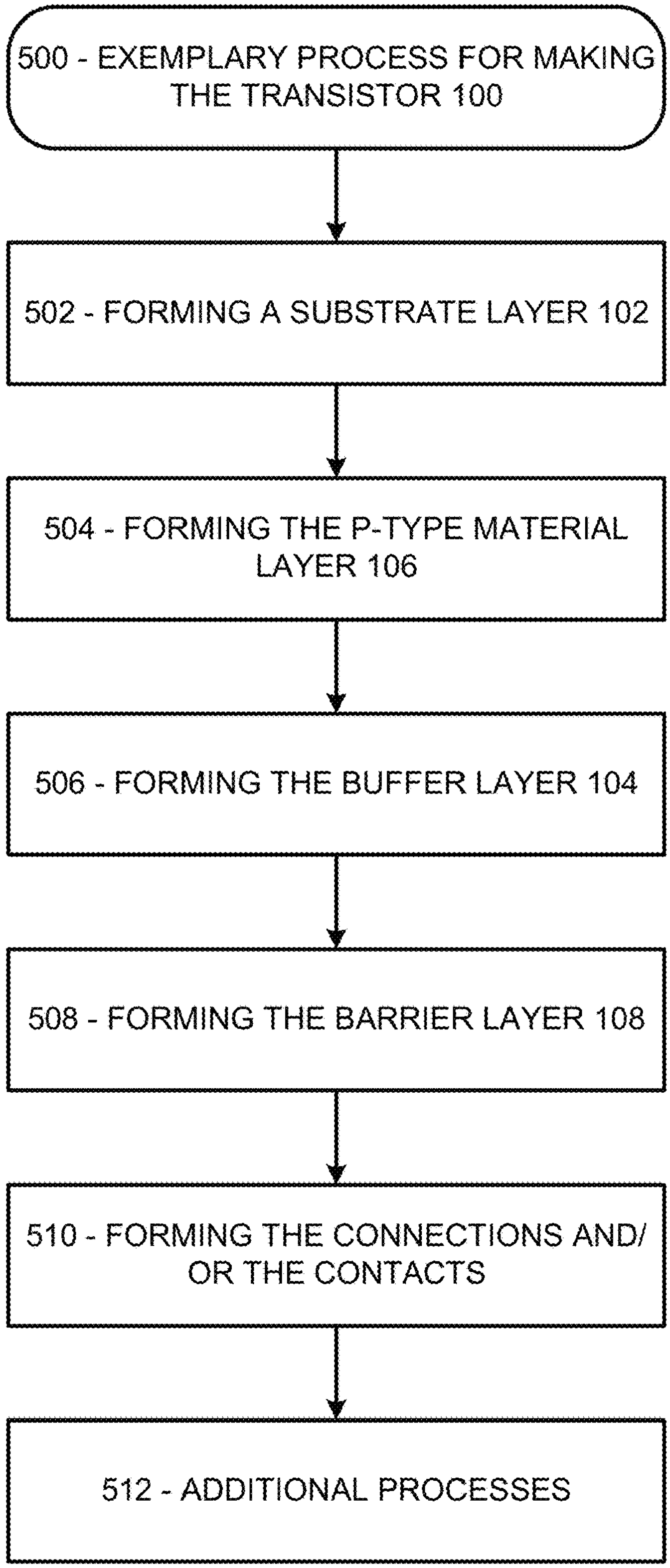
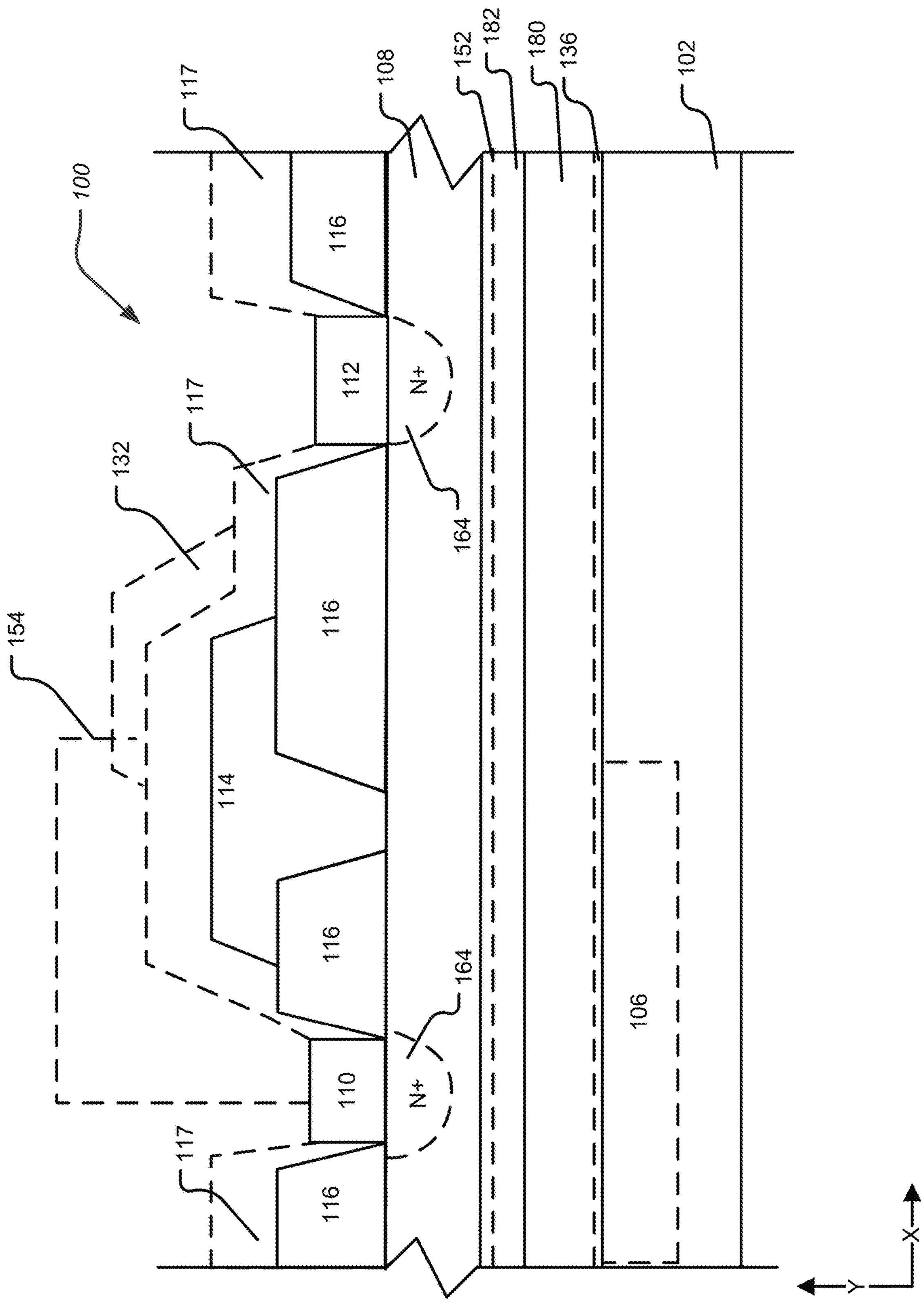

500 - EXEMPLARY PROCESS FOR MAKING THE TRANSISTOR 100
502 - FORMING A SUBSTRATE LAYER 102
504 - FORMING THE P-TYPE MATERIAL LAYER 106
506 - FORMING THE BUFFER LAYER 104
508 - FORMING THE BARRIER LAYER 108
510 - FORMING THE CONNECTIONS AND/ OR THE CONTACTS
512 - ADDITIONAL PROCESSES

FIG. 15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116
108
152
182
180
136
102
112
N+
117
132
164
116
154
114
116
164
110
N+
117
116
106
X
Y

FIG. 16

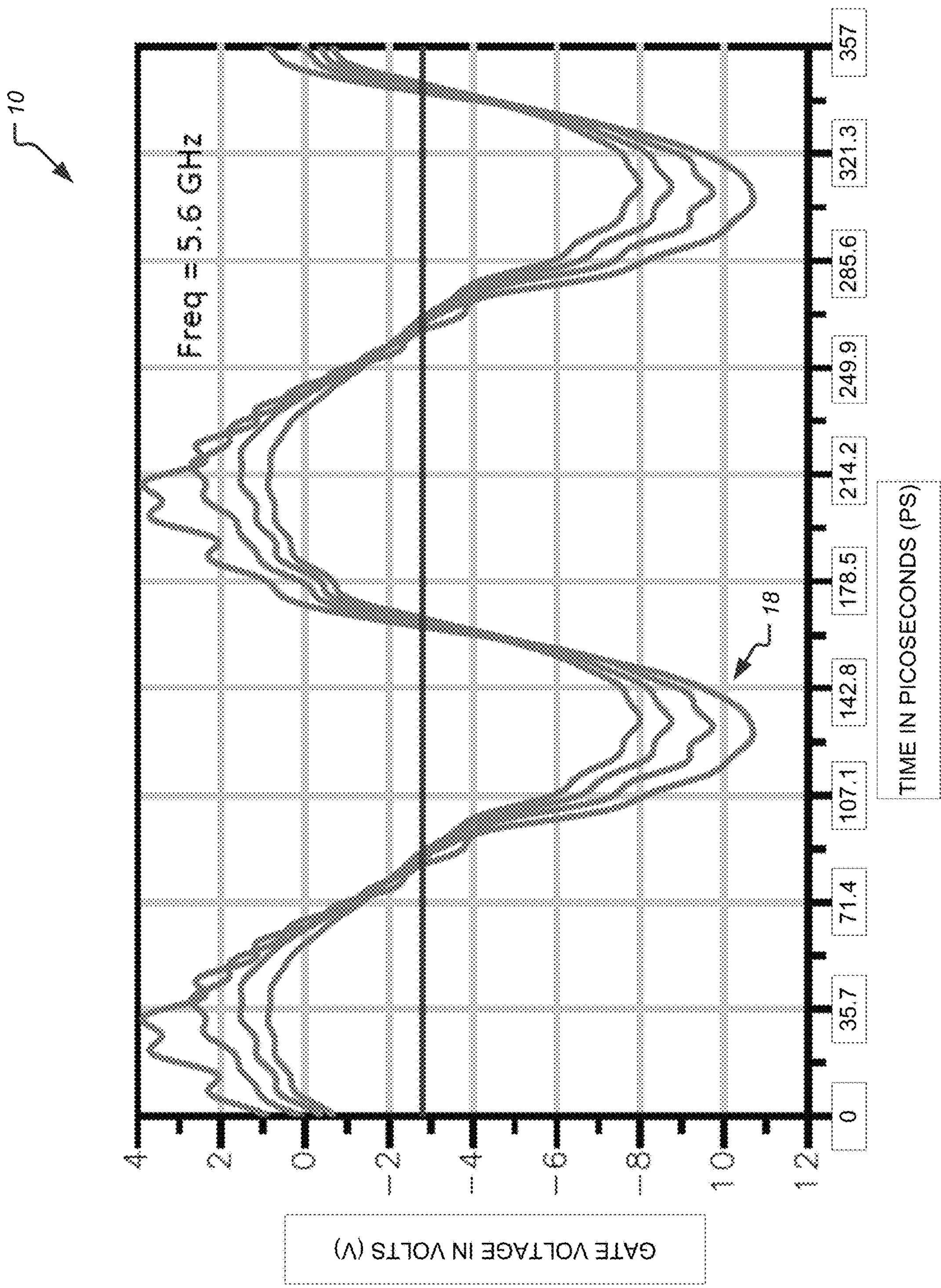
10
Freq = 5.6 GHz
18
GATE VOLTAGE IN VOLTS (V)
4
2
0
−2
−4
−6
−8
−10
−12
TIME IN PICOSECONDS (PS)
0
35.7
71.4
107.1
142.8
178.5
214.2
249.9
285.6
321.3
357

FIG. 17

CIRCUITS AND GROUP III-NITRIDE TRANSISTORS WITH BURIED P-LAYERS AND CONTROLLED GATE VOLTAGES AND METHODS THEREOF

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

This invention is related to federally sponsored research and development under Contract No.—N000164-19-C-WP50 for the Department of Defense. The invention was made with U.S. government support. The U.S. government has certain rights in the invention.

FIELD OF THE DISCLOSURE

The disclosure relates to circuits and Group III-Nitride transistors with buried p-layers and controlled gate voltages and methods thereof. The disclosure further relates to circuits and Group III-Nitride transistors with buried p-layers and controlled gate voltages. The disclosure further relates to methods of implementing, making, and/or associated with circuits and Group III-Nitride transistors with buried p-layers and controlled gate voltages. The disclosure relates to microelectronic devices and more particularly to circuits and gallium nitride high-electron mobility transistors with buried p-type layers reducing lag. The disclosure also relates to a process of implementing microelectronic devices and more particularly to a process of implementing circuits and gallium nitride high-electron mobility transistors with buried p-type layers reducing lag.

BACKGROUND OF THE DISCLOSURE

Group III-Nitride based or gallium nitride (GaN) based high-electron mobility transistors (HEMTs) are very promising candidates for high power radiofrequency (RF) applications, both in discrete and MMIC (Monolithic Microwave Integrated Circuit) forms. Current GaN HEMT designs use buffer layers that include traps to achieve desired breakdown. However, these traps cause memory effects that adversely affect performance. In particular, these designs show some trapping associated with what is called a "lag effect."

Accordingly, there is a need for a solution to addressing a lag effect and/or other negative performance issues in Group-III nitride HEMTs and improving the performance of such devices.

SUMMARY OF THE DISCLOSURE

One aspect includes an apparatus, that includes a substrate; a group III-Nitride barrier layer; a source electrically coupled to the group III-Nitride barrier layer; a gate on the group III-Nitride barrier layer; a drain electrically coupled to the group III-Nitride barrier layer; a p-region being arranged at or below the group III-Nitride barrier layer; and a gate control circuit configured to control a gate voltage of the gate, where at least a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain.

One aspect includes a method of making a device that includes providing a substrate; providing a group III-Nitride barrier layer; electrically coupling a source to the group III-Nitride barrier layer; arranging a gate on the group III-Nitride barrier layer; electrically coupling a drain to the group III-Nitride barrier layer; electrically coupling a gate control circuit to the gate to control a gate voltage of the gate; and providing a p-region being arranged at or below the group III-Nitride barrier layer, where at least a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain.

One general aspect includes a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages.

One general aspect includes a method associated with a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages.

One general aspect includes a method of implementing a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages.

One general aspect includes a method of making a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages.

Additional features, advantages, and aspects of the disclosure may be set forth or apparent from consideration of the following detailed description, drawings, and claims. Moreover, it is to be understood that both the foregoing summary of the disclosure and the following detailed description are exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the disclosure, are incorporated in and constitute a part of this specification, illustrate aspects of the disclosure and together with the detailed description serve to explain the principles of the disclosure. No attempt is made to show structural details of the disclosure in more detail than may be necessary for a fundamental understanding of the disclosure and the various ways in which it may be practiced. In the drawings:

FIG. **1** illustrates a schematic circuit diagram for a transistor package according to an aspect of the disclosure.

FIG. **2** illustrates a schematic circuit diagram for a transistor package according to an aspect of the disclosure according to FIG. **1**.

FIG. **3** illustrates simulated gate voltages for a transistor package with respect to time implementing a gate control circuit of the disclosure according to FIG. **2**.

FIG. **4** illustrates a schematic circuit diagram for a transistor package according to an aspect of the disclosure according to FIG. **1**.

FIG. **5** illustrates simulated gate voltages for transistor package with respect to time implementing a gate control circuit of the disclosure according to FIG. **4**.

FIG. **6** shows a cross-sectional view of one aspect of a transistor according to the disclosure.

FIG. **7** shows a cross-sectional view of an aspect of the transistor according to FIG. **6**.

FIG. **8** shows a cross-sectional view of an aspect of the transistor according to FIG. **6**.

FIG. **9** illustrates a semiconductor device that may include a plurality of unit cell transistors in accordance with an aspect of the disclosure.

FIG. **10** is a schematic cross-sectional view taken along line X-X of FIG. **9**.

FIG. **11** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

FIG. **12** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

FIG. **13** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

FIG. **14** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

FIG. **15** shows a process for making a transistor according to the disclosure.

FIG. **16** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

FIG. **17** illustrates simulated gate voltage for a transistor package with respect to time without implementation of a gate control circuit of the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

The aspects of the disclosure and the various features and advantageous details thereof are explained more fully with reference to the non-limiting aspects and examples that are described and/or illustrated in the accompanying drawings and detailed in the following description. It should be noted that the features illustrated in the drawings are not necessarily drawn to scale, and features of one aspect may be employed with other aspects, as the skilled artisan would recognize, even if not explicitly stated herein. Descriptions of well-known components and processing techniques may be omitted so as to not unnecessarily obscure the aspects of the disclosure. The examples used herein are intended merely to facilitate an understanding of ways in which the disclosure may be practiced and to further enable those of skill in the art to practice the aspects of the disclosure. Accordingly, the examples and aspects herein should not be construed as limiting the scope of the disclosure, which is defined solely by the appended claims and applicable law. Moreover, it is noted that like reference numerals represent similar parts throughout the several views of the drawings and in the different aspects disclosed.

It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the disclosure. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

It will be understood that when an element such as a layer, region, or substrate is referred to as being "on" or extending "onto" another element, it can be directly on or extend directly onto the another element or intervening elements may also be present. In contrast, when an element is referred to as being "directly on" or extending "directly onto" another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being "over" or extending "over" another element, it can be directly over or extend directly over the another element or intervening elements may also be present. In contrast, when an element is referred to as being "directly over" or extending "directly over" another element, there are no intervening elements present. It will also be understood that when an element is referred to as being "connected" or "coupled" to another element, it can be directly connected or coupled to the another element or intervening elements may be present. In contrast, when an element is referred to as being "directly connected" or "directly coupled" to another element, there are no intervening elements present.

Relative terms such as "below" or "above" or "upper" or "lower" or "horizontal" or "vertical" may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises," "comprising," "includes," and/or "including" when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

In addition to the type of structure, the characteristics of the semiconductor material from which a transistor is formed may also affect operating parameters. Of the characteristics that affect a transistor's operating parameters, the electron mobility, saturated electron drift velocity, electric breakdown field, and thermal conductivity may have an effect on a transistor's high frequency and high power characteristics.

Electron mobility is the measurement of how rapidly an electron is accelerated to its saturated velocity in the presence of an electric field. In the past, semiconductor materials, which had a high electron mobility, were preferred because more current could be developed with a lesser field, resulting in faster response times when a field is applied. Saturated electron drift velocity is the maximum velocity that an electron can obtain in the semiconductor material. Materials with higher saturated electron drift velocities are preferred for high frequency applications because the higher velocity translates to shorter times from source to drain.

Electric breakdown field is the field strength at which breakdown of the Schottky junction and the current through the gate of the device suddenly increases. A high electric breakdown field material is preferred for high power, high frequency transistors because larger electric fields generally can be supported by a given dimension of material. Larger electric fields allow for faster transients as the electrons can be accelerated more quickly by larger electric fields than by smaller ones.

Thermal conductivity is the ability of the semiconductor material to dissipate heat. In typical operations, all transistors generate heat. In turn, high power and high frequency transistors usually generate larger amounts of heat than small signal transistors. As the temperature of the semiconductor material increases, the junction leakage currents generally increase and the current through the field effect transistor generally decreases due to a decrease in carrier mobility with an increase in temperature. Therefore, if the heat is dissipated from the semiconductor, the material will remain at a lower temperature and be capable of carrying larger currents with lower leakage currents.

The disclosure includes both extrinsic and intrinsic semiconductors. Intrinsic semiconductors are undoped (pure). Extrinsic semiconductors are doped, meaning an agent has been introduced to change the electron and hole carrier concentration of the semiconductor at thermal equilibrium. Both p-type and n-type semiconductors are disclosed, with p-types having a larger hole concentration than electron concentration, and n-types having a larger electron concentration than hole concentration.

Silicon carbide (SiC) has excellent physical and electronic properties, which should theoretically allow production of electronic devices that can operate at higher temperatures, higher power, and higher frequency than devices produced from silicon (Si) or gallium arsenide (GaAs) substrates. The high electric breakdown field of about 4×E6 V/cm, high saturated electron drift velocity of about 2.0×E7 cm/sec and high thermal conductivity of about 4.9 W/cm-° K indicate that SiC would be suitable for high frequency and high power applications. In some aspects, the transistor of the disclosure comprises Si, GaAs or other suitable substrates.

GaN based HEMTs are very promising candidates for high power RF applications, both in discrete and MMIC forms. GaN HEMT designs may use buffer layers that include traps to achieve desired breakdown. However, these traps may cause memory effects that adversely affect performance. To overcome this limitation, structures with buried p layers may be utilized to enable obtaining breakdown with minimal trapping. These devices show reduction and/or elimination of drain lag effect and the part of trapping associated with that effect. However, they still show some trapping associated with what is called a "gate lag effect," particularly at high negative gate voltages.

In particular, according to the disclosure structures using buried p layers may be utilized to reduce trapping in GaN HEMTs. These have been shown to reduce and/or eliminate drain lag associated trapping and also reduce and/or eliminate gate lag trapping up to about −8V reverse bias on the gate. However, it is been found that gate lag trapping effects may occur when the gate voltage drops below −8V. This is undesirable and needs to be reduced or eliminated.

FIG. **17** illustrates simulated gate voltage for a transistor package with respect to time without implementation of a gate control circuit of the disclosure. In particular, FIG. **17** illustrates simulated gate voltages **10** for the transistor package with respect to time without implementation of a gate control circuit of the disclosure. In particular, the vertical axis corresponds to gate voltage for a transistor package in volts (V); and the horizontal axis corresponds to time in picoseconds (ps). Additionally, the transistor package may be implemented as an amplifier operating at 5.6 GHz with four input power levels.

More specifically, FIG. **17** shows a negative peak of −8 to −11 volts (indicated by arrow **18**) while operating at the four input power levels around the peak power-added efficiency (PAE) without implementation of the gate control circuit of the disclosure. This may not be the worst case negative voltage since impedance at the second harmonic may not explicitly controlled. In this regard, gate lag trapping effects may occur when the gate voltage drops below −8V in certain applications. This is undesirable and needs to be reduced or eliminated. For other applications, the gate voltage going below other negative voltage values may likewise induce gate lag trapping effects.

The disclosure provides a number of different gate control circuits and/or processes for reducing a negative gate voltage and/or gate voltage swing. The gate control circuits and/or processes may connect to or be incorporated in matching circuits on the input of a transistor, such as a HEMT, FET, and/or the like to reduce a negative gate voltage and/or a negative gate voltage swing. There are a number of gate control circuits and/or processes disclosed for controlling the peak negative gate voltage that may include a use of a second harmonic short at a transistor gate, such as a HEMT gate, and/or a use of a diode based circuit. In particular, use of a diode based circuit may compensate for the voltage dependent input capacitance of the device such as a HEMT device. In this regard, the gate control circuit implementations of the disclosure may reduce occurrences the gate voltage drops below −8V in certain transistor applications thereby reducing and/or limiting gate lag trapping effects. For other applications, the gate control circuit implementations of the disclosure may reduce occurrences where the gate voltage drops below other negative voltage values thereby reducing and/or limiting gate lag trapping effects.

Circuits having related components may be utilized for improving efficiency and/or linearity of a transistor. However, the disclosure configures and implements a gate control circuit differently and in particular configures and implements a gate control circuit for managing gate lag of a transistor and therefor transient response. In this regard, the gate control circuit implementations of the disclosure reduce occurrences where the gate voltage drops below −8V in certain transistor applications thereby reducing and/or limiting gate lag trapping effects. For other applications, the gate control circuit implementations of the disclosure may reduce occurrences where the gate voltage drops below other negative voltage values thereby reducing and/or limiting gate lag trapping effects. Simulation results are presented herein to show the reduction of negative gate voltage peaks and thereby improve the gate lag as well as the overall lag of the transistor application.

Additionally, a buried p layer is used in the implementations of the disclosure to reduce and/or eliminate drain lag and/or gate lag. In addition, the disclosure implements the gate control circuit to reduce negative gate voltage, reduce negative gate voltage swing, reduce gate lag, eliminate gate lag, and/or the like. In addition, the disclosure may implement the gate control circuit in conjunction with an input matching circuit to reduce negative gate voltage, reduce negative gate voltage swing, reduce gate lag, eliminate gate lag, and/or the like. Using these techniques to control gate lag is novel. Reduction and/or elimination of drain lag and gate lag offers significant competitive advantage in GaN technology area. Additionally, the proposed structures can be used in GaN discrete HEMTs, MMICs, and/or the like for use in commercial applications, defense applications, and/or the like. The combination of a buried p layer together with the gate control circuit reduces drain lag, eliminates drain lag, reduces drain lag trapping, eliminates drain lag trapping, reduces gate lag, eliminates gate lag, reduces gate lag trapping, eliminates gate lag trapping, and/or the like.

More specifically, overall lag in transistors such as GaN based HEMTs may be a combination of both gate lag effect and drain lag effect. In particular aspects, implementing various approaches to reduce a first type of lag affect may result in a second type of lag effect becoming more prevalent, increasing, and/or the like. Accordingly, addressing overall lag in transistors such as GaN based HEMTs may require implementation of structures to reduce the first type of lag effect, which results in the second type of lag effect becoming prevalent, which may require additional implementations to address the second type of lag effect.

In one aspect, overall lag in transistors such as GaN based HEMTs may be a combination of both gate lag effect and drain lag effect. In particular aspects, implementing various approaches that include buried p-layers to reduce overall lag in transistors such as GaN based HEMTs may require implementation of the gate control circuit implementations of the disclosure to reduce occurrences where the gate voltage drops below −8V in certain transistor applications thereby reducing and/or limiting gate lag trapping effects. For other applications, the gate control circuit implementations of the disclosure reduce occurrences where the gate voltage drops below other negative voltage values thereby reducing and/or limiting gate lag trapping effects.

FIG. **1** illustrates a schematic circuit diagram for a transistor package according to an aspect of the disclosure.

In particular, FIG. **1** illustrates a schematic circuit diagram for a transistor package **200** that may include a transistor **100**, an input matching network **212**, and an output matching network **216**. The transistor **100** may include a gate **114**, a drain **112**, and a source **110**. In various aspects, the transistor package **200** may be implemented as a radiofrequency package, an amplifier package, and/or the like

In the transistor **100**, lag can be a problematic side effect that is detrimental to the performance and/or operation of the transistor **100**. In particular, lag may be presented as a drain lag and/or a gate lag. Lag can impact performance of the transistor **100** and may include slower response, sluggish response, distortion, and/or the like. Moreover, lag may present with a larger impact to the transistor **100** when implemented as an amplifier due to the transient nature of the various input and output signals from the transistor **100**.

The disclosure presents a number of structural implementations to the transistor **100** that reduces the drain lag and/or gate lag as further described herein. However, these structural implementations to reduce drain lag of the transistor **100** may in some applications result in presenting gate lag to be a greater issue impacting performance of the transistor **100**. For example, these structural implementations to reduce drain lag in the transistor **100** may in some applications detrimentally impact gate lag, increase the impact of gate lag, increase gate lag, and/or the like. In particular, where the gate voltage drops below −8V in certain transistor applications may result in an increase in impact of gate lag, increase gate lag, increase in gate lag trapping effects, and/or the like. For other applications, where the gate voltage drops below other negative voltage values may result in an increase in impact of gate lag, increase gate lag, increase in gate lag trapping effects, and/or the like.

Accordingly, the disclosure further presents a configuration with the transistor package **200** and/or the transistor **100**, which may be implemented as a gate voltage control circuit **300**, gate control circuit, and/or the like (hereinafter the gate voltage control circuit **300**) to address a gate lag, an impact of gate lag, an increase of gate lag, an increase in gate lag trapping effects, and/or the like of the transistor **100**. The gate voltage control circuit **300** may be implemented to reduce gate lag in conjunction with the drain lag reducing structural configurations and gate lag reducing structural configurations of the transistor **100** to reduce the overall lag of the transistor **100**. The gate voltage control circuit **300** may shape the waveform of the gate voltage to prevent the gate voltage from reaching higher values. In particular, the gate voltage control circuit **300** may be configured to address peak gate voltages, and/or the like. More specifically, the gate voltage control circuit **300** may be configured to reduce and/or limit peak gate voltages, and/or the like for the transistor package **200** and/or the transistor **100**. Accordingly, the gate voltage control circuit **300** of the disclosure may reduce occurrences where the gate voltage drops below −8V in certain transistor applications thereby reducing and/or limiting gate lag trapping effects. For other applications, the gate voltage control circuit **300** of the disclosure may reduce occurrences where the gate voltage drops below other negative voltage values thereby reducing and/or limiting gate lag trapping effects.

Although some circuits may be implemented to address distortion, efficiency and/or the like in conjunction with the input matching circuit, the gate voltage control circuit **300** of the disclosure is distinguished from these implementations as the gate voltage control circuit **300** may be configured and/or arranged to reduce and/or limit peak gate voltages, and/or the like for the transistor package **200** and/or the transistor **100**. More specifically, the gate voltage control circuit **300** may be configured and/or arranged to reduce and/or limit peak gate voltages, and/or the like for the transistor package **200** and/or the transistor **100** in conjunction with the drain lag structural configurations and gate lag structural configurations of the transistor **100** to reduce the overall lag of the transistor package **200** and/or the transistor **100**. This is an unexpected result.

In one aspect, the gate voltage control circuit **300** may be configured with an apparent capacitance as a function of a capacitance of the transistor **100**. The gate voltage control circuit **300** may control the apparent capacitance at the gate input of the gate **114** of the transistor **100**. The gate voltage control circuit **300** may be constructed using gate material during the same manufacturing step of the transistor **100**. In particular, the gate voltage control circuit **300** may be connected between the input matching network **212** and the gate **114** of the transistor **100**. In other aspects, the gate voltage control circuit **300** may be implemented and arranged within and/or as part of the input matching network **212**.

The input matching network **212** may be connected between an RF signal input lead **214** and the gate **114** of the transistor **100**. The transistor **100** may include an output terminal that may be the drain **112**. The output terminal of the transistor **100** of the transistor package **200** may be connected to an RF output lead **218** through the output matching network **216**. The input matching network **212** and the output matching network **216** may be configured increase the usable bandwidth of the device and/or the like.

The input matching network **212** may include one or more inductances, impedances, capacitors, and/or the like connected between the RF signal input lead **214** and the transistor **100**. In some aspects, the input matching network **212** may include a plurality of input matching circuits, each of which may include one or more inductances, impedances, capacitors, and/or the like connected between the RF signal input lead **214** and the transistor **100**.

The output matching network **216** may include one or more inductances, impedances, capacitors, and/or the like connected between the RF output lead **218** and the source **110** of the transistor **100**. In some aspects, the output matching network **216** may include a plurality of output matching circuits, each of which may include one or more inductances, impedances, capacitors, and/or the like connected between the RF output lead **218** and the source **110** of the transistor **100**.

FIG. **2** illustrates a schematic circuit diagram for a package according to an aspect of the disclosure according to FIG. **1**.

In one or more aspects, the gate voltage control circuit **300** may include a second harmonic short circuit tuned to reduce and/or limit peak gate voltages, and/or the like for the transistor package **200** and/or the transistor **100**. This in conjunction with the drain lag structural configurations and/or gate lag structural configurations of the transistor **100** as described herein reduce the overall lag of the transistor package **200** and/or the transistor **100**. This is an unexpected result.

In particular, the gate voltage control circuit **300** implementing a second harmonic short circuit may include a resonant circuit, for example an LC resonator, that is configured to provide a second harmonic short at an input. Additionally, the gate voltage control circuit **300** implementing a second harmonic short circuit may be tied to ground.

FIG. **3** illustrates simulated gate voltage for a transistor package with respect to time implementing a gate control circuit of the disclosure according to FIG. **2**.

In particular, FIG. **3** illustrates simulated gate voltage **280** for a transistor package with respect to time implementing the gate voltage control circuit **300** of the disclosure in conjunction with the transistor **100** and/or the transistor package **200**. In particular, the vertical axis corresponds to gate voltage for a transistor package in volts (V); and the horizontal axis corresponds to time in picoseconds (ps). More specifically, FIG. **3** shows a negative peak of −6 to −8 volts (arrow **282**) while operating at the four input power levels around the peak power-added efficiency (PAE) with implementation of the gate voltage control circuit **300** of the disclosure. Additionally, the transistor package may be implemented as an amplifier operating at 5.6 GHz with four input power levels.

Accordingly, the FIG. **3** simulated gate voltage **280** illustrates gate voltages at less negative values with implementation of the gate voltage control circuit **300** in comparison to operation without implementation of a gate control circuit of the disclosure as illustrated in FIG. **17**. More specifically, implementation of the gate voltage control circuit **300** in the transistor **100** and/or the transistor package **200** reduces or eliminates gate lag trapping effects that may occur when the gate voltage drops below −8V in certain applications.

FIG. **4** illustrates a schematic circuit diagram for a package according to an aspect of the disclosure according to FIG. **1**.

The gate voltage control circuit **300** may implement a diode-based circuit. For example, the gate voltage control circuit **300** may implement a diode **302** as a compensation diode. The diode **302** may be configured to counteract the non-linear input capacitance of the transistor **100** to make a nearly constant net input capacitance. In one aspect, the gate voltage control circuit **300** may implement the diode **302** and the configuration of the diode **302** may be chosen to reduce gate lag in conjunction with the transistor **100** having a drain lag and/or gate lag reducing or eliminating configuration. In particular, the gate voltage control circuit **300** may implement the diode **302** together with a voltage source VCR. Moreover, the gate voltage control circuit **300** may implement the diode **302** together with a voltage source VCR that is tied to ground.

FIG. **5** illustrates simulated gate voltage for a transistor package with respect to time implementing a gate control circuit of the disclosure.

In particular, FIG. **5** illustrates simulated gate voltage **290** for a transistor package with respect to time implementing the gate voltage control circuit **300** of the disclosure in conjunction with the transistor **100** and/or the transistor package **200**. In particular, the vertical axis corresponds to gate voltage for a transistor package in volts (V); and the horizontal axis corresponds to time in picoseconds (ps). More specifically, FIG. **5** shows a negative peak of −5 to −7 volts (arrow **292**) while operating at the four input power levels around the peak power-added efficiency (PAE) with implementation of the gate voltage control circuit **300** of the disclosure. Additionally, the transistor package may be implemented as an amplifier operating at 5.6 GHz with four input power levels.

Accordingly, the FIG. **5** simulated gate voltage **290** illustrates gate voltages at less negative values with implementation of the gate voltage control circuit **300** in comparison to operation without implementation of a gate control circuit of the disclosure as illustrated in FIG. **17**. More specifically, implementation of the gate voltage control circuit **300** in the transistor **100** and/or the transistor package **200** reduces or eliminates gate lag trapping effects that may occur when the gate voltage drops below −8V in certain applications.

FIG. **6** shows a cross-sectional view of an aspect of a transistor according to the disclosure.

In particular, FIG. **6** shows a cross-sectional view of a transistor **100**. More specifically, the transistor **100** described in detail herein may be utilized in conjunction with the transistor package **200** and/or the gate voltage control circuit **300** as described above. As further described below, the transistor **100** includes structures and configurations to reduce overall lag of the transistor **100** including drain lag and gate lag in conjunction with a buried p-region or p-type material layer.

The transistor **100** may include a substrate layer **102**. The substrate layer **102** may be made of Silicon Carbide (SiC). In some aspects, the substrate layer **102** may be a semi-insulating SiC substrate, a p-type substrate, an n-type substrate, and/or the like. In some aspects, the substrate layer **102** may be very lightly doped. In one aspect, the background impurity levels may be low. In one aspect, the background impurity levels may be 1 E15/cm^3 or less. In one aspect, the substrate layer **102** may be formed of SiC selected from the group of 6H, 4H, 15R, 3C SiC, or the like. In one aspect, the substrate layer **102** may be formed of SiC that may be semi-insulating and doped with vanadium or any other suitable dopant or undoped of high purity with defects providing the semi-insulating properties.

In another aspect, the substrate layer **102** may be GaAs, GaN, or other material suitable for the applications described herein. In another aspect, the substrate layer **102** may include sapphire, spinel, ZnO, silicon, or any other material capable of supporting growth of Group III-nitride materials. In particular aspects, the substrate layer **102** may include a planer upper surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). In particular aspects, the substrate layer **102** may include a planer lower surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). Where upper and lower are defined along the Y axis.

The transistor **100** may include a buried p-region or p-type material layer **106** that may be formed within the substrate layer **102**. The p-type material layer **106** may be provided solely in the substrate layer **102**, extend from the substrate layer **102** to epitaxial layers within the transistor **100**, or located solely in the epitaxial layers of the transistor **100**. The dopants can be incorporated into the epitaxial layers by ion implantation alone, through epitaxial growth, or a combination of both. The p-type material layer **106** can span multiple layers and include multiple areas of different or graded p-doping. In accordance with other aspects of the disclosure, the p-type material layer **106** may also be formed below the barrier layer **108** between the barrier layer **108** and the substrate layer **102** and/or within the substrate layer **102**.

In accordance with aspects of the disclosure, at least some portions of the substrate layer **102** may include the p-type material layer **106**. In accordance with aspects of the disclosure, the p-type material layer **106** may be formed by ion implantation of aluminum (Al) and annealing. In other aspects, the p-type material layer **106** may be formed by ion implantation of boron, gallium, or any other material that may form a p-type layer or a combination of these. In one aspect, the p-type material layer **106** may be formed by implantation and annealing of Al prior to the growth of any GaN layers. In one aspect, the ion implementation may utilize channeling implants. In one aspect, the channeling implants may include aligning the ion beam to the substrate layer **102**. Alignment of the ion beam may result in increased implanting efficiency. In other aspects, the ion implantation may not utilize channeling.

Aspects of the disclosure may utilize implant channeling to controllably form implanted regions of the p-type material layer **106** in silicon carbide implementations of the substrate layer **102** that are highly uniform by depth and also result in reduced lattice damage. Channeling is experienced when ions are implanted along a crystal axis of the substrate layer **102**. When a direction of implantation is close to a major axis of the crystal lattice, the atoms in the crystal lattice appear to "line up" relative to the direction of implantation, and the implanted ions appear to travel down the channels created by the crystal structure to form the p-type material layer **106**. This reduces the likelihood of collisions between the implanted ions and the atoms in the crystal lattice. As a result, the depth of the implant of the p-type material layer **106** may be greatly increased.

In general, channeling occurs in silicon carbide when the direction of implantation is within about ±0.2° of a crystallographic axis of the silicon carbide crystal. In some aspects, the implantation may be greater than ±0.2° of the crystallographic axis of the silicon carbide crystal, however the implantation may be less effective. For example, when the direction of implantation is more than about ±0.2° of a crystallographic axis of the silicon carbide crystal, the atoms in the lattice may appear to be randomly distributed relative to the direction of implantation, which may reduce channeling effects. As used herein, the term "implant angle" refers to the angle between the direction of implantation and a crystallographic axis, such as the c-axis or <0001> axis, of the semiconductor layer into which ions are implanted. Thus, an implant angle of less than about 2° relative to the c-axis of a silicon carbide layer may be expected to result in channeling. However, other implant angles may be utilized as well.

In one aspect, the p-type material layer **106** may be formed by ion implantation of ^{27}Al in 4H—SiC implanted with channeling conditions with an implant energy of E_1=100 keV with a dose of 1E13 cm^2 at 25° C. In one aspect, the p-type material layer **106** may be formed by ion implantation of ^{27}Al in 4H—SiC implanted with channeling conditions with an implant energy of E_2=300 keV with a dose of 1E13 cm^2 at 25° C. However, other implant energies and doses are contemplated as well. For example, in some aspects the implant energy may be 20 keV to 80 keV, 80 keV to 120 keV, 120 keV to 160 keV, 160 keV to 200 keV, 200 keV to 240 keV, 240 keV to 280 keV, 280 keV to 340 keV, 340 keV to 400 keV, 20 keV to 400 keV, and/or 80 keV to 340 keV; and in some aspects the implant dose may be 0.6E13 cm^2 to 0.8E13 cm^2, 0.8E13 cm^2 to 1.2E13 cm^2, 1.2E13 cm^2 to 1.6E13 cm^2, 1.6E13 cm^2 to 2E13 cm^2, 0.6E13 cm^2 to 2E13 cm^2, and/or 0.8E13 cm^2 to 1.2E13 cm^2. Additionally, it should be noted that the p-type material layer **106** may be formed by implantation of other materials such as Boron (B), Gallium (Ga), and/or the like, and may be followed by a high temperature anneal.

In one aspect, the ion implantation may result in the p-type material layer **106** being a deep layer. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 1 μm or less. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.7 μm or less. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.5 μm or less. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.3 μm to 0.5 μm. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.2 μm to 0.6 μm. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.4 μm to 0.6 μm. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.6 μm to 0.8 μm. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.6 μm to 1.6 μm. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.6 μm to 2.1 μm. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 1 μm to 5 μm. In one aspect, the p-type material layer **106** implantation and/or doping may be in the range of 5E15 to 5E17 per cm^3 and extend to depths up to 5 μm.

In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.05% to 0.3% of a thickness of the substrate layer **102**. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.05% to 0.1% of a thickness of the substrate layer **102**. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.1% to 0.15% of a thickness of the substrate layer **102**. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.15% to 0.2% of a thickness of the substrate layer **102**. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.2% to 0.25% of a thickness of the substrate layer **102**. In one aspect, the ion implantation may result in the p-type material layer **106** having a thickness of 0.25% to 0.3% of a thickness of the substrate layer **102**.

The p-type material layer **106** may be implanted within the substrate layer **102** and may be subsequently annealed. Annealing may allow for the implantation to be activated. In one aspect, a masking layer material may be utilized during implantation. In some aspects, during annealing of the p-type material layer **106**, a cap layer material may be used to cover the wafer surface to prevent dissociation of the substrate at high temperatures. Once the p-type material layer **106** has been formed, the masking layer material may be removed. Annealing may be performed at a temperature range of 1500-1850° C. for 5 minutes-30 minutes. Other annealing time and temperature profiles are contemplated as well.

In some aspects, the substrate layer **102** may be made of a p-type material SiC substrate. Further in this aspect, the substrate layer **102** being a p-type material SiC substrate may be subsequently subjected to the processes as described herein including implantation of additional p-type layers. In aspects of the transistor **100** of the disclosure, the p-type material layer **106** may be neutralized to limit the length of the p-type material layer **106**. In one aspect, neutralizing may include implantation of impurities. In one aspect, neutralizing the p-type material layer **106** may include absorbing the charge of the p-type material layer **106** with a material of opposite polarity. Another way to limit the length of the p-type material layer **106** may be to etch the p-type material layer **106**. Another way to limit the length of the p-type material layer **106** may be to use a masking material to limit the area for implantation.

In aspects of the transistor **100** of the disclosure, the p-type material layer **106** may be formed by growing the p-type material layer **106**. Growth may be epitaxial, for example. To limit the length of the p-type material layer **106**, the p-type material layer **106** may be etched or otherwise neutralized. In aspects of the transistor **100** of the disclosure, the substrate layer **102** may be etched and the p-type material layer **106** may be formed by growing the p-type material layer **106**. In one aspect, the growth may be epitaxial.

In aspects of the transistor **100** of the disclosure, the p-type material layer **106** may be an epitaxial layer and may be GaN. In some aspects, the p-type material layer **106** may be an epitaxial layer and may be GaN and the p-type material layer **106** may include magnesium (Mg), carbon (C), and/or Zinc. In some aspects, the p-type material layer **106** may be an epitaxial layer and may be GaN and the p-type material layer **106** may include implantation of magnesium (Mg), carbon (C), and/or Zinc.

In aspects of the transistor **100** of the disclosure, the substrate layer **102** may be etched and the p-type material layer **106** may be formed by growing the p-type material layer **106**. In one aspect, the growth may be epitaxial.

In aspects of the transistor **100** of the disclosure, the p-type material layer **106** may be an epitaxial layer formed of SiC. In some aspects, the p-type material layer **106** may be an epitaxial layer and may be SiC and the p-type material layer **106** may include Al and/or Br. In some aspects, the p-type material layer **106** may be an epitaxial layer and may be SiC and the p-type material layer **106** may include implantation of Al and/or Br.

In some aspects, the p-type material layer **106** may be under 0.6 μm in thickness. In some aspects, the p-type material layer **106** may be under 0.5 μm in thickness. In some aspects, the p-type material layer **106** may be under 0.4 μm in thickness. In some aspects, the p-type material layer **106** may be under 0.3 μm in thickness. In some aspects, the p-type material layer **106** may be under 0.2 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.1 and 0.6 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.5 and 0.6 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.4 and 0.5 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.3 and 0.4 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.2 and 0.3 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.1 and 0.3 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.05 and 0.25 μm in thickness. In some aspects, the p-type material layer **106** may be between 0.15 and 0.25 μm in thickness.

In aspects of the transistor **100** of the disclosure, the p-type material layer **106** may be a graded layer. In one aspect, the p-type material layer **106** may be a step-graded layer. In one aspect, the p-type material layer **106** may be multiple layers. In one aspect, the p-type material layer **106** may be a graded layer. In one aspect, the p-type material layer **106** may be a step-graded layer. In one aspect, the p-type material layer **106** may be multiple layers. In particular aspects, the p-type material layer **106** may include a planer upper surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). In particular aspects, the p-type material layer **106** may include a planer lower surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). Where upper and lower are defined along the Y axis. Depending on the material of the substrate layer **102**, a nucleation layer **136** may be formed on the substrate layer **102** to reduce a lattice mismatch between the substrate layer **102** and a next layer in the transistor **100**. In one aspect, the nucleation layer **136** may be formed directly on the substrate layer **102**. In other aspects, the nucleation layer **136** may be formed on the substrate layer **102** with intervening layer(s), such as SiC epitaxial layer(s) formed on a SiC implementation of the substrate layer **102**. The nucleation layer **136** may include different suitable materials, such as a Group III-Nitride material, e.g., $Al_xIn_{y1-x-y}GaN$ (where 0<=x<=1, 0<=y<=1, x+y<=1). The nucleation layer **136** may be formed on the substrate layer **102** using known semiconductor growth techniques such as Metal Oxide Chemical Vapor Deposition (MOCVD), Hydride Vapor Phase Epitaxy (HVPE), Molecular Beam Epitaxy (MBE), or the like. In some aspects, the nucleation layer is Aluminum Nitride (AlN) or Aluminum Gallium Nitride (AlGaN), such as undoped AlN or AlGaN.

In some aspects, a buffer layer **104** may be formed directly on the nucleation layer **136** or on the nucleation layer **136** with intervening layer(s). Depending on the aspect, the buffer layer **104** may be formed of different suitable materials such as a Group III-nitride such as $Al_xGa_yIn_{(1-x-y)}N$ (where 0<=x<=1, 0<=y<=1, x+y<=1), e.g., GaN, AlGaN, AlN, and the like, or another suitable material. In one aspect, the buffer layer **104** is formed of GaN. The buffer layer **104** or portions thereof may be doped with dopants, such as, Fe and/or C or alternatively can be wholly or partly undoped. In one aspect, the buffer layer **104** is directly on the substrate layer **102**. In particular aspects, the buffer layer **104** may include a planer upper surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). In particular aspects, the buffer layer **104** may include a planer lower surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). Where upper and lower are defined along the Y axis.

In one aspect, the buffer layer **104** may include an upper portion of high purity GaN and the buffer layer **104** may also include a lower portion that may form an AlGaN back barrier to achieve better electron confinement. In one aspect, the lower portion that forms the back barrier may be AlGaN of n type. The back barrier construction may be implemented in any of the aspects of the disclosure.

In one aspect, the buffer layer **104** may be high purity GaN. In one aspect, the buffer layer **104** may be high purity GaN that may be a low-doped n-type. In one aspect, the buffer layer **104** may also use a higher band gap Group III-nitride layer as a back barrier, such as an AlGaN back barrier, on the other side of the buffer layer **104** from a barrier layer **108** to achieve better electron confinement.

In one aspect, the buffer layer **104** may have a buffer layer thickness defined as a distance between an upper surface of the substrate layer **102** and a lower surface of the barrier layer **108**. In one aspect, the buffer layer thickness may be less than 0.8 microns, less than 0.7 microns, less than 0.6 microns, less than 0.5 microns, or less than 0.4 microns. In one aspect, the buffer layer thickness may have a range of 0.8 microns to 0.6 microns, 0.7 microns to 0.5 microns, 0.6 microns to 0.4 microns, 0.5 microns to 0.3 microns, 0.4 microns to 0.2 microns, or 0.7 microns to 0.3 microns.

In one aspect, the transistor **100** may have an intervening layer(s) thickness defined as a length between an upper surface of the substrate layer **102** and a lower surface of the barrier layer **108**. In one aspect, the intervening layer(s) thickness may be less than 0.8 microns, less than 0.7 microns, less than 0.6 microns, less than 0.5 microns, or less than 0.4 microns. In one aspect, the intervening layer(s) thickness may have a range of 0.8 microns to 0.6 microns, 0.7 microns to 0.5 microns, 0.6 microns to 0.4 microns, 0.5 microns to 0.3 microns, or 0.4 microns to 0.2 microns.

The barrier layer **108** may be formed on the buffer layer **104**. In one aspect, the barrier layer **108** may be formed directly on the buffer layer **104**, and in other aspects, the barrier layer **108** is formed on the buffer layer **104** with intervening layer(s). Depending on the aspect, the buffer layer **104** may be formed of different suitable materials such as a Group III-nitride such as $Al_xGa_yIn_{(1-x-y)}N$ (where 0<=x<=1, 0<=y<=1, x+y<=1), e.g., AlGaN, AlN, or InAlGaN, or another suitable material. In one aspect, the barrier layer **108** may be AlGaN, and in another aspect the barrier layer **108** is AlN. In one aspect, the barrier layer **108** may be undoped. In one aspect, the barrier layer **108** may be doped. In one aspect, the barrier layer **108** may be an n-type material. In some aspects, the barrier layer **108** may have multiple layers of n-type material having different carrier concentrations. In one aspect, the barrier layer **108** may be a Group III-nitride or a combination thereof. In particular aspects, the barrier layer **108** may include a planer upper surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). In particular aspects, the barrier layer **108** may include a planer lower surface that is generally parallel to an X axis as illustrated in FIG. **6** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). Where upper and lower are defined along the Y axis.

In one aspect, a bandgap of the buffer layer **104** may be less than a bandgap of the barrier layer **108** to form a two-dimensional electron gas (2DEG) at a heterointerface **152** between the buffer layer **104** and barrier layer **108** when biased at an appropriate level. In one aspect, a bandgap of the buffer layer **104** that may be GaN may be less than a bandgap of the barrier layer **108** that may be AlGaN to form the two-dimensional electron gas (2DEG) at a heterointerface **152** between the buffer layer **104** and barrier layer **108** when biased at an appropriate level.

In aspects of the disclosure, the heterointerface **152** may be between the barrier layer **108** and the buffer layer **104**. In one aspect, the source **110** and the drain **112** electrodes may be formed making ohmic contacts such that an electric current flows between the source **110** and the drain **112** electrodes via a two-dimensional electron gas (2DEG) induced at the heterointerface **152** between the buffer layer **104** and barrier layer **108** when the gate **114** electrode is biased at an appropriate level. In one aspect, the heterointerface **152** may be in the range of 0.005 μm to 0.007 μm, 0.007 μm to 0.009 μm, and 0.009 μm to 0.011 μm.

In one aspect, the source **110**, the drain **112** and the gate **114** may be formed on the barrier layer **108**. The source **110**, the drain **112**, and/or the gate **114** may be arranged directly on the barrier layer **108** or may be on intervening layer(s) on the barrier layer **108**, such as an AlGaN layer on an AlN barrier layer. Other or additional intervening layers are possible. For example, a spacer layer **116** of SiN, AlO, SiO, SiO_2, AlN, or the like or combinations thereof can be provided on the barrier layer **108** or other intervening layers. In one aspect, the barrier layer **108** may include a region **164** under the source **110** and/or the drain **112** that is a N+ material. In one aspect, the barrier layer **108** may include a region **164** under the source **110** and/or drain **112** that is Si doped. In one aspect, the n-type dopants in the region **164** are implanted.

In one aspect, the source **110**, the drain **112** and the gate **114** may be formed on the buffer layer **104**. The source **110**, the drain **112**, and/or the gate **114** may be arranged directly on the buffer layer **104** or may be on intervening layer(s) on the buffer layer **104**, such as an AlGaN layer on an AlN barrier layer. In one aspect, the buffer layer **104** may include a region **164** under the source **110** and/or the drain **112** that is a N+ material. In one aspect, the buffer layer **104** may include a region **164** under the source **110** and/or drain **112** that is Si doped. In one aspect, the n-type dopants in the region **164** are implanted.

In some aspects, the source **110** and the drain **112** may be symmetrical with respect to the gate **114**. In some switch device application aspects, the source **110** and the drain **112** may be symmetrical with respect to the gate **114**. In some aspects, the source **110** and the drain **112** may be asymmetrical with respect to the gate **114**. In one aspect, the gate **114** may be a T-shaped gate. In one aspect, the gate **114** may be a non-T shaped gate.

To protect and separate the gate **114** and the drain **112**, a spacer layer **116** may be arranged on the barrier layer **108**, on a side opposite the buffer layer **104**, adjacent the gate **114**, the drain **112** and the source **110**. The spacer layer **116** may be a passivation layer made of SiN, AlO, SiO, SiO_2, AlN, or the like, or a combination incorporating multiple layers thereof. In one aspect, the spacer layer **116** is a passivation layer made of SiN. In one aspect, the spacer layer **116** can be deposited using MOCVD, plasma chemical vapor deposition (CVD), hot-filament CVD, or sputtering. In one aspect, the spacer layer **116** may include deposition of Si_3N_4. In one aspect, the spacer layer **116** forms an insulating layer. In one aspect, the spacer layer **116** forms an insulator. In one aspect, the spacer layer **116** may be a dielectric. In one aspect, a spacer layer **116** may be provided on the barrier layer **108**. In one aspect, the spacer layer **116** may include non-conducting material such as a dielectric. In one aspect, the spacer layer **116** may include a number of different layers of dielectrics or a combination of dielectric layers. In one aspect, the spacer layer **116** may be many different thicknesses, with a suitable range of thicknesses being approximately 0.05 to 2 microns. In one aspect, the spacer layer **116** may include a material such as a Group III nitride material having different Group III elements such as alloys of Al, Ga, or In, with a suitable spacer layer material being $Al_xIn_yGa_{1-x-y}$ (where 0<=x<=1 and 0<=y<=1, x+y<=1).

In some aspects, the gate **114** may be deposited in a channel formed in the spacer layer **116**, and a T-gate may be formed using semiconductor processing techniques understood by those of ordinary skill in the art. Other gate configurations are possible.

In aspects of the transistor **100** of the disclosure, the substrate layer **102** may be silicon carbide and include a carbon face. In one aspect, the substrate layer **102** may be silicon carbide and include a carbon face arranged adjacent the buffer layer **104**. In one aspect, the substrate layer **102** may be silicon carbide and include a carbon face and the substrate layer **102** may be flipped so as to be arranged adjacent the buffer layer **104**. In this aspect, the buffer layer **104** may be GaN having a nitrogen face adjacent the carbon face of the substrate layer **102**. In one aspect, the buffer layer **104** may be GaN having alternating GaN and N layers with a N layer and/or a nitrogen face adjacent the carbon face of the substrate layer **102**.

In aspects of the transistor **100** of the disclosure, the buffer layer **104** may include nonpolar GaN. In one aspect, the buffer layer **104** may include semipolar GaN. In one aspect, the buffer layer **104** may include hot wall epitaxy. In one aspect, the buffer layer **104** may include hot wall epitaxy having a thickness in the range of 0.15 microns to 0.25 microns, 0.2 microns to 0.3 microns, 0.25 microns to 0.35 microns, 0.3 microns to 0.35 microns, 0.35 microns to 0.4 microns, 0.4 microns to 0.45 microns, 0.45 microns to 0.5 microns, 0.5 microns to 0.55 microns, or 0.15 microns to 0.55 microns. The p-type material layer **106** may help avoid breakdowns and problems with material impurities. For example, without a p-type material layer **106**, the transistor **100** may need impurities, which do not discharge well. The p-type material layer **106** may be formed beneath the gate **114**, and may extend toward the source **110** and the drain **112** of the device.

In aspects of the transistor **100** of the disclosure, the buffer layer **104** may be designed to be of the high purity type where the Fermi level is in the upper half of the bandgap, which reduces slow trapping effects normally observed in GaN HEMTs. In this regard, the traps under the Fermi level are filled always and thus slow transients may be prevented. In some aspects, the buffer layer **104** may be as thin as possible consistent with achieving good crystalline quality. Applicants have already demonstrated 0.4 μm layers with good quality.

In aspects of the transistor **100** of the disclosure, a $Al_xIn_yGa_{1-x-y}$ (where 0<=x<=1 and 0<=y<=1, x+y<=1) nucleation layer **136** or buffer layer **104** may be grown on the substrate layer **102** via an epitaxial crystal growth method, such as MOCVD (Metalorganic Chemical Vapor Deposition), HVPE (Hydride Vapor Phase Epitaxy) or MBE (Molecular Beam Epitaxy). The formation of the nucleation layer **136** may depend on the material of the substrate layer **102**.

In aspects of the transistor **100** of the disclosure, the buffer layer **104** may be formed with Lateral Epitaxial Overgrowth (LEO). LEO can, for example, improve the crystalline quality of GaN layers. When semiconductor layers of a HEMT are epitaxial, the layer upon which each epitaxial layer is grown may affect the characteristics of the device. For example, LEO may reduce dislocation density in epitaxial GaN layers.

With reference to the description of FIG. **11**, the transistor **100** may include a second spacer layer **117** that may be formed on the spacer layer **116** and the gate **114**. With reference to the description of FIG. **12**, the transistor **100** may include a field plate **132**. With reference to the description of FIG. **13**, the transistor **100** may include a connection **154** to the field plate **132**.

FIG. **7** shows a cross-sectional view of an aspect of the transistor according to FIG. **6**.

In one aspect of the disclosure, the p-type material layer **106** may not extend over the entire area of the transistor **100**. In this regard, the p-type material layer **106** may be selectively arranged as described herein, the p-type material layer **106** may be arranged over the entire length and selectively removed as described herein, the p-type material layer **106** may be arranged over the entire length and selectively electrically neutralized as described herein, or the like. Accordingly, the specific constructions of the p-type material layer **106** described below encompass any of these processes that result in the p-type material layer **106** having an operating construction and arrangement as noted below. In other words, the length and/or size of the p-type material layer **106** does not include a part that is partially electrically neutralized, partially etched, or the like. The length and/or size of the p-type material layer **106** may depend on the application of the transistor **100**, requirements for the transistor **100**, and the like. Limiting a length of the p-type material layer **106** reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

As shown in FIG. **7**, the p-type material layer **106** may be present in limited areas as described in further detail below. In some aspects, the p-type material layer **106** may be present in a gate-source region. In some aspects, the p-type material layer **106** may be present in a gate-source region and also partly under the gate **114**. In some aspects, the p-type material layer **106** may be arranged at least partially under the gate **114** and/or the source **110**. In some aspects, the p-type material layer **106** may be arranged at least partially under the gate **114** and/or and not arranged under the source **110**.

In one aspect, the p-type material layer **106** may be arranged at least partially vertically under the gate **114** along the y-axis and may extend along the x-axis partially toward the source **110** and the drain **112**. In this aspect, no portion of the p-type material layer **106** may be located vertically along the y-axis below the source **110**; and no portion of the p-type material layer **106** may be located vertically along the y-axis below the source **110**. In this aspect, a portion of the substrate layer **102** may be free of the p-type material layer **106** on a source side of the transistor **100**; and a portion of the substrate layer **102** may be free of the p-type material layer **106** on a drain side of the transistor **100**. In this regard, a source side of the transistor **100** is defined as a side of the transistor **100** extending from the gate **114** toward and past the source **110** as illustrated in FIG. **7**; and a drain side of the transistor **100** is defined as a side of the transistor **100** extending from the gate **114** toward and past the drain **112** as illustrated in FIG. **7**.

In one aspect, the p-type material layer **106** may be arranged at least partially vertically under the gate **114** along the y-axis and may extend along the x-axis partially toward the source **110** and the drain **112**. In this aspect, only a portion of the p-type material layer **106** may be located vertically along the y-axis below the source **110**; and no portion of the p-type material layer **106** may be located vertically along the y-axis below the source **110**. In this aspect, a portion of the substrate layer **102** may not include the p-type material layer **106** located vertically along the y-axis below the source **110**. In this aspect, a portion of the substrate layer **102** may be free of the p-type material layer **106** on a source side of the transistor **100**; and a portion of the substrate layer **102** may be free of the p-type material layer **106** on a drain side of the transistor **100**.

In one aspect, the p-type material layer **106** may be arranged at least partially vertically under the gate **114** along the y-axis and may extend along the x-axis partially toward the source **110** and the drain **112**. In this aspect, a portion of the p-type material layer **106** may be located vertically along the y-axis entirely below the source **110**; and no portion of the p-type material layer **106** may be located vertically along the y-axis below the drain **112**. In this aspect, a portion of the substrate layer **102** may not include the p-type material layer **106** located vertically along the y-axis past the source **110**. In this aspect, a portion of the substrate layer **102** may be free of the p-type material layer **106** on a source side of the transistor **100**; and a portion of the substrate layer **102** may be free of the p-type material layer **106** on a drain side of the transistor **100**.

In one aspect, the p-type material layer **106** may be arranged vertically under the gate **114** along the y-axis and may extend along the x-axis partially toward the source **110** and the drain **112**. In this aspect, a portion of the p-type material layer **106** may be located vertically along the y-axis entirely below the source **110**; and no portion of the p-type material layer **106** may be located vertically along the y-axis below the source **110**. In this aspect, a portion of the substrate layer **102** may not include the p-type material layer **106** located vertically along the y-axis past the source **110**. In this aspect, a portion of the substrate layer **102** may be free of the p-type material layer **106** on a source side of the transistor **100**; and a portion of the substrate layer **102** may be free of the p-type material layer **106** on a drain side of the transistor **100**.

With reference to FIG. **7**, various dimensions of components of the transistor **100** will be described in order to define dimensions of the p-type material layer **106**. The gate **114** may have a width LG along a lower surface of the gate **114** that is adjacent the barrier layer **108** that is parallel to the X axis. In particular, the width LG may extend from one lower corner of the gate **114** to the other lower corner of the gate **114**. The definition of the width LG is illustrated in FIG. **7**. In some aspects, the width LG may be between 0.05 μm and 0.6 μm, 0.5 μm and 0.6 μm, 0.4 μm and 0.5 μm, 0.3 μm and 0.4 μm, 0.2 μm and 0.3 μm, 0.1 μm and 0.2 μm, or 0.1 μm and 0.05 μm in length along the x-axis. In some aspects, a width of the gate **114** above a lower surface may be greater than the width LG as illustrated in FIG. **7**.

The distance from the gate **114** to the source **110** may be defined as distance LGS. In particular, the distance LGS may be defined as a distance from a lower corner of the gate **114** on a source side to a lower corner of the source **110** on a gate side. The definition of the distance LGS is illustrated in FIG. **7**.

The distance from the gate **114** to the drain **112** may be defined as the distance LGD. In particular, the distance LGD may be defined as a distance from a lower corner of the gate **114** on a drain side to a lower corner of the drain **112** on a gate side. The definition of the distance LGD is illustrated in FIG. **7**.

In one aspect, the p-type material layer **106** may extend laterally along the x-axis from at least beneath the lower corner of the gate **114** on a source side toward the source **110** a distance LGPS. The definition of the distance LGPS is illustrated in FIG. **7**. In some aspects, the distance LGPS may be between 1 μm and 6 μm, 5 μm and 6 μm, 4 μm and 5 μm, 3 μm and 4 μm, 2 μm and 3 μm, or 1 μm and 3 μm in length along the x-axis.

In one aspect, the p-type material layer **106** may extend laterally along the x-axis from at least beneath the lower corner of the gate **114** on a drain side toward the drain **112** a distance LGPD. In some aspects, the distance LGPD may be between 0.1 μm and 0.6 μm, 0.5 μm and 0.6 μm, 0.4 μm and 0.5 μm, 0.3 μm and 0.4 μm, 0.2 μm and 0.3 μm, or 0.1 μm and 0.3 μm in length along the x-axis.

Accordingly, a length of the p-type material layer **106** may be a sum of the distance LGPD, the width LG, and the distance LGPS. In this regard, a length of the p-type material layer **106** reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

In one aspect, the length LGPS may be 100% to 700% of LG, 100% to 200% of LG, 200% to 300% of LG, 300% to 400% of LG, 400% to 500% of LG, 500% to 600% of LG, or 600% to 700% of LG.

In one aspect, the length LG may be 10% to 180% of LGPD, 10% to 20% of LGPD, 20% to 30% of LGPD, 30% to 40% of LGPD, 40% to 50% of LGPD, 50% to 60% of LGPD, 60% to 70% of LGPD, 70% to 80% of LGPD, 80% to 90% of LGPD, 90% to 100% of LGPD, 100% to 110% of LGPD, 110% to 120% of LGPD, 110% to 130% of LGPD, 130% to 140% of LGPD, 140% to 150% of LGPD, 150% to 160% of LGPD, 160% to 170% of LGPD, or 170% to 180% of LGPD.

In one aspect, the length LGS may be 10% to 180% of LGPS, 10% to 20% of LGPS, 20% to 30% of LGPS, 30% to 40% of LGPS, 40% to 50% of LGPS, 50% to 60% of LGPS, 60% to 70% of LGPS, 70% to 80% of LGPS, 80% to 90% of LGPS, 90% to 100% of LGPS, 100% to 110% of LGPS, 110% to 120% of LGPS, 110% to 130% of LGPS, 130% to 140% of LGPS, 140% to 150% of LGPS, 150% to 160% of LGPS, 160% to 170% of LGPS, or 170% to 180% of LGPS.

In one aspect, the length LG may be 10% to 180% of LGPD, 10% to 20% of LGPD, 20% to 30% of LGPD, 30% to 40% of LGPD, 40% to 50% of LGPD, 50% to 60% of LGPD, 60% to 70% of LGPD, 70% to 80% of LGPD, 80% to 90% of LGPD, 90% to 100% of LGPD, 100% to 110% of LGPD, 110% to 120% of LGPD, 110% to 130% of LGPD, 130% to 140% of LGPD, 140% to 150% of LGPD, 150% to 160% of LGPD, 160% to 170% of LGPD, or 170% to 180% of LGPD.

In one or more aspects, a part of a source side of the substrate layer **102** may be free of the p-type material layer **106**. In one or more aspects, a part of a drain side of the substrate layer **102** may be free of the p-type material layer **106**. In one or more aspects, a part of a source side of the substrate layer **102** may be free of the p-type material layer **106** and a part of a drain side of the substrate layer **102** may be free of the p-type material layer **106**. In one or more aspects, the p-type material layer **106** may be arranged under and across a length of the gate **114** and may extend toward the source **110** and the drain **112**.

In one or more aspects, a distance LGD may be a distance from a lower corner of the gate **114** on the drain **112** side to a lower corner of the drain **112** on a gate side; a distance LGS may be a distance from a lower corner of the gate **114** on the source **110** side to a lower corner of the source **110** on a gate side; and the distance LGD may be greater than the distance LGS. In one or more aspects, a distance LGPS may define a length of a portion of the p-type material layer **106** from a lower corner of the gate **114** on the source **110** side toward the source **110**; a distance LGPD may define a length of a portion of the p-type material layer **106** from a lower corner of the gate **114** on the drain **112** side toward the drain **112**; and the distance LGPS may be equal to the distance LGPD. In one or more aspects, a distance LGPS may define a length of a portion of the p-type material layer **106** from a lower corner of the gate **114** on the source **110** side toward the source **110**; a distance LGPD may define a length of a portion of the p-type material layer **106** from a lower corner of the gate **114** on the drain **112** side toward the drain **112**; and the distance LGPS may be greater than the distance LGPD. In one or more aspects, a distance LGPS may define a length of a portion of the p-type material layer **106** from a lower corner of the gate **114** on the source **110** side toward the source **110**; a distance LGPD may define a length of a portion of the p-type material layer **106** from a lower corner of the gate **114** on the drain **112** side toward the drain **112**; and the distance LGPD may be greater than the distance LGPS.

In one or more aspects, the p-type material layer **106** may extend toward the source **110** but does not vertically overlap the source **110**. In one or more aspects, the p-type material layer **106** may vertically overlap the source **110**. In one or more aspects, the p-type material layer **106** may extend toward the drain **112** but does not vertically overlap the drain **112**. In one or more aspects, the p-type material layer **106** may vertically overlap the drain **112**. In one or more aspects, the p-type material layer **106** may be electrically connected to the gate **114**. In one or more aspects, the gate **114** may be electrically connected to any external circuit or voltage. In one or more aspects, the p-type material layer **106** may have no direct electrical connections. In one or more aspects, the p-type material layer **106** may be electrically connected to the source **110**.

In some aspects, part of the voltage from a drain **112** to a source **110** may be dropped in the p-type material layer **106** region. This may also deplete the channel in the lateral direction. The lateral depletion may reduce the lateral field and increase breakdown voltage. Alternatively, a more compact structure can be obtained for a required breakdown voltage. The p-type material layer **106** may eliminate the need to have C or Fe doping of the buffer needed to sustain the applied drain voltage. Elimination of C and Fe leads to decreased current reduction under operating conditions (no trapping). Moreover, in some aspects the p-type material layer **106** may support the field.

In some aspects, the p-type material layer **106** may also be configured to have a varying doping and/or implantation profile perpendicular to the surface. In some aspects, the p-type material layer **106** may also be configured to have a varying profile perpendicular to the surface extending into the cross-sectional views of the Figures. The profile may be configured to achieve desired breakdown voltage, device size, switching time, and the like.

FIG. **8** shows a cross-sectional view of an aspect of the transistor according to FIG. **6**.

In one aspect, the p-type material layer **106** may not extend over the entire area of the substrate layer **102** as shown by the arrow LENGTH P as shown in FIG. **8**. In this regard, the p-type material layer **106** may be selectively arranged as described in detail below, the p-type material layer **106** may be arranged over the entire length and selectively removed as described in detail below, the p-type material layer **106** may be arranged over the entire length and selectively electrically neutralized as described in detail below, or the like. Accordingly, the specific constructions of the p-type material layer **106** described below encompass any of these configurations that result in the p-type material layer **106** having an operating construction and arrangement as noted below. In other words, the length and/or size of the p-type material layer **106** does not include a part that is partially electrically neutralized or partially etched. The length and/or size of the p-type material layer **106** may depend on the application of the transistor **100**, requirements for the transistor **100**, and the like.

With reference to the aspects further described below, the p-type material layer **106** may extend horizontally along the X axis parallel to the arrow LENGTH P. Moreover, the p-type material layer **106** may extend horizontally parallel to the arrow LENGTH P to a point defined by a line that is perpendicular (parallel to the y-axis) to the arrow LENGTH P and extends through a component of the transistor **100** as illustrated.

In one aspect, of the disclosure, the p-type material layer **106** may extend laterally from at least beneath the source **110** to a position beneath a first edge **124** of the gate **114**. In particular, the first edge **124** may be an edge of the gate **114** on a side of the gate **114** adjacent the drain **112** and which may also be a lowest surface of the gate **114**.

In certain aspects of the disclosure, the p-type material layer **106** may extend to a point within about 0 to about 0.7 μm of a first edge **124** of the gate **114**. In certain aspects of the disclosure, the p-type material layer **106** may extend to a point within about 0 to about 0.5 μm of the first edge **124** of the gate **114**. In certain aspects of the disclosure, the p-type material layer **106** may extend to a point within about 0 to about 0.3 μm of the first edge **124** of the gate **114**.

In one aspect, of the disclosure, the p-type material layer **106** may extend laterally from at least beneath the source **110** to a position beneath a second edge **122** of the gate **114**. In particular, the second edge **122** may be an edge of the gate **114** on a side of the gate **114** adjacent the source **110** and which may also be a lowest surface of the gate **114**.

In certain aspects of the disclosure, the p-type material layer **106** may extend to a point within about 0 to about 0.7 μm of the second edge **122** of the gate **114**. In certain aspects of the disclosure, the p-type material layer **106** may extend to a point within about 0 to about 0.5 μm of the second edge **122** of the gate **114**. In certain aspects of the disclosure, the p-type material layer **106** may extend to a point within about 0 to about 0.3 μm of the second edge **122** of the gate **114**.

In other aspects, a length of the p-type material layer **106** LENGTH P can also be seen in relation to positions and/or lengths of other components based on the length SD as illustrated in FIG. **8**. The length SD in this case may be the length between an edge **142** of the source **110** toward an edge **144** of the drain **112** as shown in FIG. **8**. In particular, the edge **142** may be defined as an edge or surface on the source **110** that is parallel to the Y axis on a side of the source **110** opposite to the gate **114**; and, the edge **144** may be defined as an edge or surface on the drain **112** that is parallel to the Y axis on a side of the drain **112** opposite to the gate **114**.

In one aspect, the length of the p-type material layer **106** may extend from 10% to 20% of the length of SD, meaning the p-type material layer **106** may extend 10% to 20% past the edge **142** of the source **110** toward the drain **112**. In one aspect, the length of the p-type material layer **106** may extend from 20% to 30% of the length of SD, meaning the p-type material layer **106** may extend 20% to 30% past the edge **142** of the source **110** toward the drain **112**. In one aspect, the length of the p-type material layer **106** may extend from 30% to 40% of the length of SD, meaning the p-type material layer **106** may extend 30% to 40% past the edge **142** of the source **110** toward the drain **112**. In one aspect, the length of the p-type material layer **106** may extend from 40% to 50% of the length of SD, meaning the p-type material layer **106** may extend 40% to 50% past the edge **142** of the source **110** toward the drain **112**. In one aspect, the length of the p-type material layer **106** may extend from 50% to 60% of the length of SD, meaning the p-type material layer **106** may extend 50% to 60% past the edge **142** of the source **110** toward the drain **112**. In one aspect, the length of the p-type material layer **106** may extend from 60% to 70% of the length of SD, meaning the p-type material layer **106** may extend 60% to 70% past the edge **142** of the source **110** toward the drain **112**. In one aspect, the length of the p-type material layer **106** may extend from 70% to 80% of the length of SD, meaning the p-type material layer **106** may extend 70% to 80% past the edge **142** of the source **110** toward the drain **112**.

FIG. **9** illustrates a semiconductor device that may include a plurality of unit cell transistors in accordance with an aspect of the disclosure.

As shown in FIG. **9**, aspects of the disclosure may include a semiconductor device **400** that may include a plurality of the transistor **100**. In particular, the transistor **100** may be one of a plurality of unit cells **430** implemented in the semiconductor device **400**.

In particular, FIG. **9** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **9** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **9** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

The semiconductor device **400** may include a gate bus **402** that may be connected to a plurality of gate fingers **406** that may extend in parallel in a first direction (e.g., the Z-direction indicated in FIG. **9**) that connect to or form part of the gate **114**. A source bus **410** may be connected to a plurality of parallel ones of source contacts **416** that connect to or form part of the source **110**. In some aspects, the source bus **410** may be connected to a ground voltage node on an underside of the semiconductor device **400**. A drain bus **420** may be connected to a plurality of drain contacts **426** that connect to or form part of the drain **112**.

As can be seen in FIG. **9**, each gate finger **406** may run along the Z-direction between a pair of adjacent ones of the source contact **416** and the drain contact **426**. The semiconductor device **400** may include a plurality of unit cells **430**, where each one of the plurality of unit cells **430** includes an implementation of the transistor **100**. One of the plurality of unit cells **430** is illustrated by the dashed Box in FIG. **9**, and includes a gate finger **406** that extends between adjacent ones of the source contact **416** and the drain contact **426**.

The "gate width" refers to the distance by which the gate finger **406** overlaps with its associated one of the source contact **416** and drain contact **426** in the Z-direction. That is, "width" of a gate finger **406** refers to the dimension of the gate finger **406** that extends in parallel to and adjacent an implementation of the source contact **416** and the drain contact **426** (the distance along the Z-direction). Each of the plurality of unit cells **430** may share one of the source contact **416** and/or the drain contact **426** with one or more adjacent ones of the plurality of unit cells **430**. Although a particular number of the of the plurality of unit cells **430** is illustrated in FIG. **9**, it will be appreciated that the semiconductor device **400** may include more or less of the plurality of unit cells **430**.

FIG. **10** is a schematic cross-sectional view taken along line X-X of FIG. **9**.

Referring to FIG. **10**, the semiconductor device **400** may include a semiconductor structure **440** that includes the substrate layer **102**, the buffer layer **104**, the barrier layer **108**, and/or the like as described herein. The source contact **416** and the drain contact **426** may be on the barrier layer **108** as described herein. The gate fingers **406** may be on the substrate layer **102** between the source contacts **416** and the drain contacts **426** as described herein. While the gate fingers **406**, the source contact **416**, and the drain contacts **426** are all shown schematically in FIG. **9** and FIG. **10** as having a similar "dimension," it will be appreciated that each may have different shapes and dimensions consistent with the disclosure.

FIG. **11** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

In particular, FIG. **11** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **11** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **11** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like. In particular, FIG. **11** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **11** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **11** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

FIG. **11** further illustrates implementation of the second spacer layer **117**. The second spacer layer **117** may be provided over the gate **114** and/or the spacer layer **116**. The second spacer layer **117** may be a passivation layer made of SiN, AlO, SiO, SiO_2, AlN, or the like, or a combination incorporating multiple layers thereof.

In one aspect, the second spacer layer **117** is a passivation layer made of SiN. In one aspect, the second spacer layer **117** can be deposited using MOCVD, plasma chemical vapor deposition (CVD), hot-filament CVD, or sputtering. In one aspect, the second spacer layer **117** may include deposition of Si_3N_4. In one aspect, the second spacer layer **117** forms an insulating layer. In one aspect, the second spacer layer **117** forms an insulator. In one aspect, the second spacer layer **117** may be a dielectric. In one aspect, a second spacer layer **117** may be provided on the spacer layer **116**. In one aspect, the second spacer layer **117** may include non-conducting material such as a dielectric. In one aspect, the second spacer layer **117** may include a number of different layers of dielectrics or a combination of dielectric layers. In one aspect, the second spacer layer **117** may be many different thicknesses, with a suitable range of thicknesses being approximately 0.05 to 2 microns. In one aspect, the second spacer layer **117** may include a material such as a Group III nitride material having different Group III elements such as alloys of Al, Ga, or In, with a suitable spacer layer material being $Al_xIn_yGa_{1-x-y}$ (where $0<=x<=1$ and $0<=y<=1$, $x+y<=1$).

FIG. **12** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

In particular, FIG. **12** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. The transistor **100** of FIG. **12** may include the p-type material layer **106** as described above. In particular, FIG. **12** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **12** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **12** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

FIG. **12** further illustrates implementation of the field plate **132**. In one aspect, the field plate **132** may be arranged on the second spacer layer **117** between the gate **114** and drain **112**. In one aspect, the field plate **132** may be deposited on the second spacer layer **117** between the gate **114** and the drain **112**. In one aspect, the field plate **132** may be electrically connected to one or more other components in the transistor **100**. In one aspect, the field plate **132** may not be electrically connected to any other components of the transistor **100**. In some aspects, the field plate **132** may be adjacent the gate **114** and a second spacer layer **117** of dielectric material may be included at least partially over the gate **114** to isolate the gate **114** from the field plate **132**. In some aspects, the field plate **132** may overlap the gate **114** and a second spacer layer **117** of dielectric material may be included at least partially over the gate **114** to isolate the gate **114** from the field plate **132**.

The field plate **132** may extend different distances from the edge of the gate **114**, with a suitable range of distances being approximately 0.1 to 2 microns. In some aspects, the field plate **132** may include many different conductive materials with a suitable material being a metal, or combinations of metals, deposited using standard metallization methods. In one aspect, the field plate **132** may include titanium, gold, nickel, titanium/gold, nickel/gold, or the like.

In one aspect, the field plate **132** may be formed on the second spacer layer **117** between the gate **114** and the drain **112**, with the field plate **132** being in proximity to the gate **114** but not overlapping the gate **114**. In one aspect, a space between the gate **114** and field plate **132** may be wide enough to isolate the gate **114** from the field plate **132**, while being small enough to maximize a field effect provided by the field plate **132**.

In certain aspects, the field plate **132** may reduce a peak operating electric field in the transistor **100**. In certain aspects, the field plate **132** may reduce the peak operating electric field in the transistor **100** and may increase the breakdown voltage of the transistor **100**. In certain aspects, the field plate **132** may reduce the peak operating electric field in the transistor **100** and may reduce trapping in the transistor **100**. In certain aspects, the field plate **132** may reduce the peak operating electric field in the transistor **100** and may reduce leakage currents in the transistor **100**.

In other aspects, for example, the spacer layer **116** is formed on the barrier layer **108** and on the gate **114**. In such aspects, the field plate **132** can be formed directly on the spacer layer **116**. Other multiple field plate configurations are possible with the field plate **132** overlapping or non-overlapping with the gate **114** and/or multiple field plates **132** being used.

FIG. **13** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

In particular, FIG. **13** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **13** may include the p-type material layer **106** as described above. FIG. **13** further illustrates implementation of the field plate **132** that may be electrically connected to the source **110** through a connection **154**. Additionally or alternatively, the field plate **132** may be connected to the gate **114** through a connection (gate-field plate interconnect (not shown)). In particular, FIG. **13** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **13** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **13** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

In one aspect, the connection **154** may be formed on the spacer layer **116** and/or the second spacer layer **117** to extend between the source **110** and the field plate **132**. In some aspects, the connection **154** may include a conductive material, many different conductive materials, a suitable material being a metal, or combinations of metals, deposited using standard metallization methods. In one aspect, the materials may include one or more of titanium, gold, nickel, or the like.

In one aspect, the gate-field plate interconnect may be formed on the spacer layer **116** and/or the second spacer layer **117** to extend between the gate **114** and the field plate **132**. In some aspects, the gate-field plate interconnect may include a conductive material, many different conductive materials, a suitable material being a metal, or combinations of metals, deposited using standard metallization methods. In one aspect, the materials may include one or more of titanium, gold, nickel, or the like.

In particular, the transistor **100** of FIG. **13** illustrates the field plate **132** connected to the source **110** through the connection **154** (source-field plate interconnect). In one aspect, the connection **154** may be formed on the spacer layer **116** and/or the second spacer layer **117** to extend between the field plate **132** and the source **110**. In one aspect, the connection **154** may be formed with the field plate **132** during the same manufacturing step. In one aspect, a plurality of the connection **154** and/or a plurality of gate-field plate interconnect may be used. In one aspect, a plurality of the field plates **132** may be used. In one aspect, a plurality of the field plates **132** may be used and each of the plurality of field plates **132** may be stacked with a dielectric material therebetween. In some aspects, the connection **154** and/or the gate-field plate interconnect may include a conductive material, many different conductive materials, a suitable material being a metal, or combinations of metals, deposited using standard metallization methods. In one aspect, the materials may include one or more of titanium, gold, nickel, or the like.

In one aspect of the transistor **100** described herein, the gate **114** may be formed of platinum (Pt), nickel (Ni), and/or gold (Au), however, other metals known to one skilled in the art to achieve the Schottky effect, may be used. In one aspect, the gate **114** may include a Schottky gate contact that may have a three-layer structure. Such a structure may have advantages because of the high adhesion of some materials. In one aspect, the gate **114** may further include an overlayer of highly conductive metal. In one aspect, the gate **114** may be configured as a T-shaped gate.

In one aspect of the transistor **100** described herein, one or more metal overlayers may be provided on one or more of the source **110**, the drain **112**, and the gate **114**. The overlayers may be Au, Silver (Ag), Al, Pt, Ti, Si, Ni, Al, and/or Copper (Cu).

In one aspect of the transistor **100** described herein, a second buffer layer may be deposited or grown on a first implementation of the buffer layer **104** on a side of the first implementation of the buffer layer **104** opposite of the substrate layer **102**. In one aspect, the second buffer layer may be formed directly on the first implementation of the buffer layer **104**. In one aspect, the second buffer layer may be a high-purity material such as Gallium Nitride (GaN), AlN, or the like. In one aspect, the second buffer layer may be a high-purity GaN. In one aspect, the second buffer layer may be a high-purity AlN. The second buffer layer may be a p-type material or n-type material. In another aspect, the second buffer layer may be undoped.

FIG. **14** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

In particular, FIG. **14** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **14** may include the p-type material layer **106** as described above. FIG. **14** further illustrates implementation of the field plate **132** that may be electrically connected to the source **110** through a connection **154**. In particular, FIG. **14** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **14** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **14** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

In various aspects of the disclosure, the p-type material layer **106** of the transistor **100** may be buried within the substrate layer **102** and otherwise may not be electrically connected to any portion of the transistor **100**. In one aspect as illustrated in FIG. **14**, the transistor **100** may include a p-type material contact **118** that may be electrically connected to receive an external signal, bias, and/or the like. The p-type material contact **118** may be electrically connected and arranged in the substrate layer **102**, the p-type material layer **106**, the substrate layer **102**, the buffer layer **104**, the barrier layer **108**, and/or the like. The p-type material contact **118** may be formed in a recess **119** in the substrate layer **102**, the p-type material layer **106**, the substrate layer **102**, the buffer layer **104**, the barrier layer **108**, and/or the like. The recess **119** may extend down to the p-type material layer **106** to allow for the p-type material contact **118** to be created there. The recess **119** may be formed by etching, and may also use a material to define the recess **119**. The material may be removed after the recess **119** has been created.

In particular, the recess **119** may remove any material above the p-type material layer **106** within a portion of a region associated with the source **110**, exposing the p-type material layer **106** on a side opposite of the substrate layer **102**. In another aspect of the disclosure, to create a place for the p-type material contact **118**, a recess **119** may be created by removing at least part of the substrate layer **102**, the p-type material layer **106**, the substrate layer **102**, the buffer layer **104**, the barrier layer **108**, and/or the like.

In certain embodiments, the source **110** may be electrically connected to the p-type material contact **118** through a connection **138**. In certain embodiments, the field plate **132** may be electrically connected to the source **110** through the connection **154**. In certain embodiments, the field plate **132** may be connected to the source **110**, and the source **110** may be connected to the p-type material contact **118** through the connection **138**.

In certain embodiments, the gate **114** may be electrically connected to the p-type material contact **118** through a connection (not shown). In certain embodiments, the field plate **132** may be electrically connected to the gate **114** through the connection. In certain embodiments, the field plate **132** may be connected to the gate **114**, and the gate **114** may be connected to the p-type material contact **118** through the connection.

FIG. **14** is meant to broadly describe different embodiments of the present invention (e.g., different p-layer and/or field plate configurations), but in the interest of clarity, not all embodiments are expressly depicted. It should be understood that the transistor **100** structures of the present invention can be utilized with a variety of p-type material layer **106** structures as described herein and otherwise. In certain embodiments, the p-type material layer **106** structure may be electrically connected to a separate bias voltage/control signal, electrically connected to the source **110** or electrically connected to the gate **114** or not electrically connected to the source **110**, the gate **114**, and the separate bias/control signal. Such electrical connection can be through a via in the epitaxial material and/or an electrical connection outside and/or at an edge of the epitaxial material. For example, the via may be structured within the recess **119**. The p-type material layer **106** can be formed or structured in any of the different variations described herein or otherwise. Depending on the embodiment, various field plate **132** configurations are possible. For example, field plates **132** may be integral with the gate **114**, single or multiple field plates **132** are possible with or without intervening dielectric spacer layers between the field plates **132**. A field plate **132** can vertically overlap or not vertically overlap with the gate **114** or an underlying field plate **132**. The field plates **132** may be electrically connected to the gate **114** or the source **110** or one or more field plates **132** connected to the gate **114**, one or more filed plates **132** connected to the source **110**, and/or one or more field plates **132** connected to neither the source **110** nor the gate **114**.

FIG. **15** shows a process for making a transistor according to the disclosure.

In particular, FIG. **15** shows an exemplary process **500** for making the transistor **100** of the disclosure. It should be noted that the process **500** is merely exemplary and may be modified consistent with the various aspects disclosed herein. In particular, the process **500** may include any one or more aspects of the disclosure described herein. In particular, the process **500** may include making the p-type material layer **106** as described above. In this regard, the process **500** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

The process **500** may begin at step **502** by forming a substrate layer **102**. The substrate layer **102** may be formed consistent with the disclosure. For example, the substrate layer **102** may be made of Silicon Carbide (SiC). In some aspects, the substrate layer **102** may be a semi-insulating SiC substrate, a p-type substrate, an n-type substrate, and/or the like. In some aspects, the substrate layer **102** may be very lightly doped. In one aspect, the background impurity levels may be low. In one aspect, the background impurity levels may be 1E15/cm^3 or less. The substrate layer **102** may be formed of SiC selected from the group of 6H, 4H, 15R, 3C SiC, or the like. In another aspect, the substrate layer **102** may be GaAs, GaN, or other material suitable for the applications described herein. In another aspect, the substrate layer **102** may include sapphire, spinel, ZnO, silicon, or any other material capable of supporting growth of Group III-nitride materials.

The process **500** may include a step **504** of forming the p-type material layer **106**. The p-type material layer **106** may be formed as described in the disclosure. This may include implanting Al into the substrate layer **102** to form the p-type material layer **106** in the substrate layer **102**. For example, the p-type material layer **106** may be formed by ion implantation of Al and annealing. In one aspect, the p-type material layer **106** may be formed by implantation and annealing of Al prior to the growth of any GaN layers. In one aspect, the ion implantation may utilize channeling implants. In one aspect, the channeling implants may include aligning the ion beam to the substrate layer **102**. Alignment of the ion beam may result in increased implantation efficiency. In some aspects, the process **500** may further include implanting Al into the substrate layer **102** to form the p-type material layer **106** in the substrate layer **102**. Thereafter, the substrate layer **102** may be annealed as defined herein. In one aspect, the p-type material layer **106** may be formed by ion implantation of ^{27}Al in 4H—SiC implanted with channeling conditions with an implant energy of E_1=100 keV with a dose of 1E13 cm^2 at 25° C. In one aspect, the p-type material layer **106** may be formed by ion implantation of ^{27}Al in 4H—SiC implanted with channeling conditions with an implant energy of E_2=300 keV with a dose of 1E13 cm^2 at 25° C. However, other implant energies and doses are contemplated as well.

The process **500** may include a step **506** of forming the buffer layer **104** on the substrate layer **102**. The buffer layer **104** may be grown or deposited on the substrate layer **102** as described in the disclosure. In one aspect, the buffer layer **104** may be GaN. In another aspect, the buffer layer **104** may be formed with LEO. In one aspect, a nucleation layer **136** may be formed on the substrate layer **102** and the buffer layer **104** may be formed at step **506** on the nucleation layer **136**. The buffer layer **104** may be grown or deposited on the nucleation layer **136**. In one aspect, the buffer layer **104** may be GaN. In another aspect, the buffer layer **104** may be formed with LEO.

Further during the process **500** as part of step **508**, the barrier layer **108** may be formed on the buffer layer **104**. The barrier layer **108** may be formed as described in the disclosure. For example, the barrier layer **108** may be an n-type conductivity layer or may be undoped. In one aspect, the barrier layer **108** may be AlGaN.

Further during the process **500** as part of step **510**, to create a place for contact with the p-type material layer **106**, a recess may be created by removing at least part of the barrier layer **108** and at least part of the buffer layer **104**. The recess formation process may remove any material above the p-type material layer **106** within a portion of a region associated with the source **110**, exposing the p-type material layer **106** on a side opposite of the substrate layer **102**.

Further during the process **500** as part of step **512**, the source **110** may be arranged on the barrier layer **108**. The source **110** may be an ohmic contact of a suitable material that may be annealed. For example, the source **110** may be annealed at a temperature of from about 500° C. to about 800° C. for about 2 minutes. However, other times and temperatures may also be utilized. Times from about 30 seconds to about 10 minutes may be, for example, acceptable. In some aspects, the source **110** may include Al, Ti, Si, Ni, and/or Pt. In one aspect, a region **164** under the source **110** that is a N+ material may be formed in the barrier layer **108**. In one aspect, a region **164** under the drain **112** may be Si doped.

Further during the process **500** as part of step **512**, the drain **112** may be arranged on the barrier layer **108**. Like the source **110**, the drain **112** may be an ohmic contact of Ni or another suitable material, and may also be annealed in a similar fashion. In one aspect, an n+ implant may be used in conjunction with the barrier layer **108** and the contacts are made to the implant. In one aspect, a region **164** under the drain **112** that is a N+ material may be formed in the barrier layer **108**. In one aspect, a region **164** under the drain **112** may be Si doped.

Further during the process **500** as part of step **512**, the gate **114** may be arranged on the barrier layer **108** between the source **110** and the drain **112**. A layer of Ni, Pt, AU, or the like may be formed for the gate **114** by evaporative deposition or another technique. The gate structure may then be completed by deposition of Pt and Au, or other suitable materials. In some aspects, the contacts of the gate **114** may include Al, Ti, Si, Ni, and/or Pt.

Further during the process **500** as part of step **512**, the spacer layer **116** may be formed. The spacer layer **116** may be a passivation layer, such as SiN, AlO, SiO, SiO_2, AlN, or the like, or a combination incorporating multiple layers thereof, which may be deposited over the exposed surface of the barrier layer **108**.

The source **110** and the drain **112** electrodes may be formed making ohmic contacts such that an electric current flows between the source **110** and drain **112** electrodes via a two-dimensional electron gas (2DEG) induced at the heterointerface **152** between the buffer layer **104** and barrier layer **108** when a gate **114** electrode is biased at an appropriate level. In one aspect, the source **110** may be electrically coupled to the barrier layer **108**, the drain **112** may be electrically coupled to the barrier layer **108**, and the gate **114** may be electrically coupled to the barrier layer **108** such that an electric current flows between the source **110** and the drain **112** via a two-dimensional electron gas (2DEG) induced at the heterointerface **152** between the buffer layer **104** and the barrier layer **108** when the gate **114** electrode is biased at an appropriate level. In one aspect, the source **110** may be electrically coupled to the transistor **100**, the drain **112** may be electrically coupled to the transistor **100**, and the gate **114** may be electrically coupled to the transistor **100** such that an electric current flows between the source **110** and the drain **112** via a two-dimensional electron gas (2DEG) induced at the heterointerface **152** between the buffer layer **104** and the barrier layer **108** when a gate **114** is biased at an appropriate level. In various aspects, the gate **114** may control a flow of electrons in the 2DEG based on a signal and/or bias placed on the gate **114**. In this regard, depending on a composition of the layers and/or a doping of the layers, the transistor **100** can be normally on or the transistor **100** can be normally off with no bias or signal on the gate. In one aspect, the heterointerface **152** may be in the range of 0.005 μm to 0.007 μm, 0.007 μm to 0.009 μm, and 0.009 μm to 0.011 μm.

The gate **114** may extend on top of a spacer or the spacer layer **116**. The spacer layer **116** may be etched and the gate **114** deposited such that the bottom of the gate **114** is on the surface of barrier layer **108**. The metal forming the gate **114** may be patterned to extend across spacer layer **116** so that the top of the gate **114** forms a field plate **132**.

Further during some aspects of the process **500** as part of step **512**, a second spacer layer **117** may be formed and a field plate **132** may be arranged on top of the second spacer layer **117** and may be separated from the gate **114**. In one aspect, the field plate **132** may be deposited on the second spacer layer **117** between the gate **114** and the drain **112**. In some aspects, the field plate **132** may include many different conductive materials with a suitable material being a metal, or combinations of metals, deposited using standard metallization methods. In one aspect, the field plate **132** may include titanium, gold, nickel, titanium/gold, nickel/gold, or the like.

In one aspect, the connection **154** may be formed with the field plate **132** during the same manufacturing step (see FIG. **14**). In one aspect, a plurality of the field plates **132** may be used. In one aspect, a plurality of the field plates **132** may be used and each of the plurality of field plates **132** may be stacked with a dielectric material therebetween. In one aspect, the field plate **132** extends toward the edge of gate **114** towards the drain **112**. In one aspect, the field plate **132** extends towards the source **110**. In one aspect, the field plate **132** extends towards the drain **112** and towards the source

110. In another aspect, the field plate **132** does not extend toward the edge of gate **114**. Finally, the structure may be covered with a dielectric spacer layer such as silicon nitride. The dielectric spacer layer may also be implemented similar to the spacer layer **116**. Moreover, it should be noted that the cross-sectional shape of the gate **114**, shown in the Figures is exemplary. For example, the cross-sectional shape of the gate **114** in some aspects may not include the T-shaped extensions. Other constructions of the gate **114** may be utilized.

Further during some aspects of the process **500** as part of step **512**, the connection **154** may be formed. In some aspects, the field plate **132** may be electrically connected to the source **110** with the connection **154**. In one aspect, the connection **154** may be formed on the second spacer layer **117** to extend between the field plate **132** and the source **110**.

It should be noted that the steps of process **500** may be performed in a different order consistent with the aspects described above. Moreover, the process **500** may be modified to have more or fewer process steps consistent with the various aspects disclosed herein. In one aspect of the process **500**, the transistor **100** may be implemented with only the p-type material layer **106**. In one aspect of the process **500**, the transistor **100** may be implemented with the p-type material layer **106** and the p-type material layer **106**. In one aspect of the process **500**, the transistor **100** may be implemented with only the p-type material layer **106**.

FIG. **16** shows a cross-sectional view of another aspect of a transistor according to the disclosure.

In particular, FIG. **16** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. The transistor **100** of FIG. **16** may include the p-type material layer **106** as described above. In particular, FIG. **16** illustrates a transistor **100** that may include any one or more aspects of the disclosure described herein. In particular, the transistor **100** of FIG. **16** may include the p-type material layer **106** as described above. In this regard, the transistor **100** of FIG. **16** implements a length of the p-type material layer **106** as described herein that reduces gate lag effect, avoids adverse effects on RF performance for certain transistor applications, and/or the like.

Further, FIG. **16** illustrates the transistor **100** implementing a back barrier layer **180** may be formed directly on the nucleation layer **136** or on the nucleation layer **136** with intervening layer(s). In aspects of the transistor **100** of the disclosure, a back barrier layer **180** may be formed directly on the substrate layer **102** or on the substrate layer **102** with intervening layer(s). In particular, the back barrier layer **180** may be configured at least in part as a gate lag reduction structure, a gate lag elimination structure, and/or the like. In particular, the back barrier layer **180** configured at least in part as a gate lag reduction structure, a gate lag elimination structure, and/or the like in conjunction with the p-type material layer **106** configured at least in part as a drain lag reduction structure, a drain lag elimination structure, and/or the like may operate together in a synergistic manner to reduce overall lag of the transistor **100**. As further described herein, this synergistic overall reduction of lag of the transistor **100** was an unexpected result of the combined structures of the back barrier layer **180** and the p-type material layer **106**.

More specifically, the transistor **100** in conjunction with the p-type material layer **106** and the back barrier layer **180** as disclosed, associated structures thereof, and/or associated processes thereof, may provide a systematic approach to reducing lag. More specifically, the transistor **100** of the disclosure may implement the p-type material layer **106** and/or processes thereof as a drain lag reduction structure and/or process to reduce the drain lag effect and the transistor **100** of the disclosure may implement the back barrier layer **180** and/or processes thereof as a gate lag reduction structure and/or process to reduce the gate lag effect.

In this regard, it has been determined that impurities such as silicon, oxygen, carbon, and/or the like in the back barrier layer **180** may increase gate lag. In particular, that impurities provide trapping, leaking, and/or the like. More specifically, aspects of the disclosure may implement the back barrier layer **180** with low background impurity levels. In one aspect, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels. In this regard, impurities have been found to build complexes with dislocations, such as point defects, which also act as deep trap levels.

More specifically, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels where low background impurity levels may be defined as impurities less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm. Moreover, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels of silicon, oxygen, carbon, and/or the like where low background impurity levels of silicon, oxygen, carbon, and/or the like may be defined as impurities of silicon, oxygen, carbon, and/or the like less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm.

Moreover, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels of silicon and oxygen where low background impurity levels of silicon and oxygen may be defined as impurities of silicon, oxygen and carbon less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm. In one aspect, low background impurity levels of silicon and oxygen where low background impurity levels of silicon and oxygen may be defined as impurities of silicon and oxygen less than 1E16. Moreover, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels of carbon where low background impurity levels of carbon may be defined as impurities of silicon, oxygen and carbon less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm. In one aspect, low background impurity levels of carbon where low background impurity levels of carbon may be defined as impurities of carbon less than 5E16.

Additionally or alternatively, low background impurity levels may be defined as impurities between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm. In particular, low background impurity levels may be defined as impurities of silicon, oxygen, carbon, and/or the like between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm.

In particular, low background impurity levels may be defined as impurities of silicon and oxygen between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm.

In particular, low background impurity levels may be defined as impurities of silicon and oxygen between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm. In particular, low background impurity levels may be defined as impurities of carbon between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm.

Additionally, the back barrier layer **180** may be configured to provide a sharp interface to the channel layer **182**. This interface may function as a barrier for electrons. In aspects of the transistor **100** of the disclosure, the back barrier layer **180** may be a graded layer. In one aspect, the back barrier layer **180** may be a step-graded layer. In one aspect, the back barrier layer **180** may be multiple layers.

In particular aspects, the back barrier layer **180** may be a low Al concentration AlGaN buffer layer to provide a barrier to reduce electron injection into the buffer layer. In this regard, the barrier to reduce electron injection into the buffer layer results in a gate lag reduction structure, a gate lag elimination structure, and/or the like. For example, the back barrier layer **180** may be implemented with about 4% Al concentration AlGaN to provide a barrier to reduce electron injection into the buffer. In this regard, about may be within 0.5%, 1%, 1.5%, or 2%. In particular aspects, the back barrier layer **180** may be implemented with AlGaN with an Al concentration of 1% to 6%, 1% to 1.5%, 1.5% to 2%, 2% to 2.5%, 2.5% to 3%, 3% to 3.5%, 3.5% to 4%, 3.5% to 4.5%, 3.8% to 4.2%, 4% to 4.5%, 4.5% to 5%, 5% to 5.5%, or 5.5% to 6%, to provide a barrier to reduce electron injection into the buffer, a gate lag reduction structure, a gate lag elimination structure, and/or the like.

In aspects, the transistor **100** may have limited gate lag during a limited operational envelope. For example, up to about −8V (volts) reverse bias on the gate for certain implementations of a GaN HEMT. However, the back barrier layer **180** may be configured as a gate lag reduction, a gate lag elimination, and/or the like for implementations outside the limited operational envelope where gate lag trapping effects may be present. For example, implementations of the transistor **100** where the gate voltage goes below −8V. In particular, the back barrier layer **180** configured at least in part as a gate lag reduction structure, a gate lag elimination structure, and/or the like in conjunction with the p-type material layer **106** configured at least in part as a drain lag reduction structure, a drain lag elimination structure, and/or the like operate together in a synergistic manner to reduce overall lag of the transistor **100** during such low gate voltage conditions. As further described herein, this synergistic overall reduction of lag of the transistor **100** was an unexpected result of the combined structures of the back barrier layer **180** and the p-type material layer **106**.

Additionally, the back barrier layer **180** of the transistor **100** may be further configured and/or processed to reduce and/or limit gate lag effect by implementing epitaxial growth thereof. In particular, the back barrier layer **180** of the transistor **100** may be further configured and/or processed to reduce and/or limit gate lag effect by implementing epitaxial growth while reducing incorporation of background impurities such as silicon (Si), oxygen (O), carbon (C), and/or the like in the AlGaN of implementations of the back barrier layer **180**. More specifically, aspects of the disclosure may implement the back barrier layer **180** with low background impurity levels. In one aspect, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels. In this regard, impurities have been found to build complexes with dislocations, such as point defects, which also act as deep trap levels. In this regard, incorporation in the back barrier layer **180** of a high concentration of high level background impurities has been found to be a problem when using AlGaN for the back barrier layer **180**. The back barrier layer **180** may implement epitaxial growth with impurity incorporation significantly reduced. More specifically, the back barrier layer **180** may be implemented with epitaxial growth of AlGaN reducing incorporation of background impurities such as silicon (Si), oxygen (O), carbon (C), and/or the like.

More specifically, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels where low background impurity levels may be defined as impurities less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm. Moreover, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels of silicon, oxygen, carbon, and/or the like where low background impurity levels of silicon, oxygen, carbon, and/or the like may be defined as impurities of silicon, oxygen, carbon, and/or the like less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm.

Moreover, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels of silicon and oxygen where low background impurity levels of silicon and oxygen may be defined as impurities of silicon, oxygen and carbon less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm. In one aspect, low background impurity levels of silicon and oxygen where low background impurity levels of silicon and oxygen may be defined as impurities of silicon and oxygen less than 1E16. Moreover, the disclosure may implement AlGaN for the back barrier layer **180** with low background impurity levels of carbon where low background impurity levels of carbon may be defined as impurities of silicon, oxygen and carbon less than 1E17 per cubic cm. (centimeter), less than 5E16 per cubic cm., less than 1E16 per cubic cm., or less than 1E15 per cubic cm. In one aspect, low background impurity levels of carbon where low background impurity levels of carbon may be defined as impurities of carbon less than 5E16.

Additionally or alternatively, low background impurity levels may be defined as impurities between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm. In particular, low background impurity levels may be defined as impurities of silicon, oxygen, carbon, and/or the like between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm.

In particular, low background impurity levels may be defined as impurities of silicon and oxygen between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm.

In particular, low background impurity levels may be defined as impurities of silicon and oxygen between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm. In particular, low background impurity levels may be defined as impurities of carbon between 1E15 per cubic cm. and 1E17 per cubic cm., 1E15 per cubic cm. and 1E16 per cubic cm., 1E16 per cubic cm. and 5E16 per cubic cm., or 5E16 per cubic cm. and 1E17 per cubic cm.

In this regard, it has been discovered that a source of defects may be impurities, which may act as non-intentional doping and create trap centers and/or the like in the transistor **100**. To prevent deep penetration of electrons in the GaN buffer of the transistor **100** or the channel layer **182** of the transistor **100**, the back barrier layer **180** may be implemented as an AlGaN buffer as described herein and may be used to confine electrons in the channel layer **182** close to the back barrier layer **180**. The disclosed implementation and configuration of the back barrier layer **180** has additionally proven to improve break-down voltage in the transistor **100** and/or GaN HEMT implementations of the transistor **100**.

Accordingly, the transistor **100** may include the p-type material layer **106** as described herein in order for drain lag to be greatly reduced and/or eliminated. However, the transistor **100** may still suffer from the gate lag effect. For example, the transistor **100** may still suffer from the gate lag effect at elevated negative gate voltages. Traps in the buffer of the transistor **100** may be a cause this delay. Accordingly, the back barrier layer **180** may be implemented with AlGaN with very low background impurity levels of Carbon, Silicon, Oxygen, and/or the like grown on the p-type material layer **106** to dramatically improve the electron confinement and reduce and/or eliminate gate lag as well as may reduce and/or eliminate overall lag.

In particular aspects, the back barrier layer **180** may include a planar upper surface that is generally parallel to an X axis as illustrated in FIG. **16** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). In particular aspects, the back barrier layer **180** may include a planar lower surface that is generally parallel to an X axis as illustrated in FIG. **16** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). Where upper and lower are defined along the Y axis.

In some aspects, a channel layer **182** may be formed directly on the back barrier layer **180** or on the back barrier layer **180** with intervening layer(s). In one aspect, the channel layer **182** is formed of GaN.

Depending on the aspect, the channel layer **182** may be formed of different suitable materials such as a Group III-nitride such as $Al_xGa_yIn_{(1-x-y)}N$ (where 0<=x<=1, 0<=y<=1, x+y<=1), e.g., GaN, AlGaN, AlN, and the like, or another suitable material. The channel layer **182** or portions thereof may be doped with dopants, such as, Fe and/or C or alternatively can be wholly or partly undoped.

In particular aspects, the channel layer **182** may include a planar upper surface that is generally parallel to an X axis as illustrated in FIG. **16** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). In particular aspects, the channel layer **182** may include a planar lower surface that is generally parallel to an X axis as illustrated in FIG. **16** and/or is generally parallel to an Z axis (perpendicular to the X axis and the Y axis). Where upper and lower are defined along the Y axis.

In one aspect, the channel layer **182** may be high purity GaN. In one aspect, the channel layer **182** may be high purity GaN that may be a low-doped n-type. In one aspect, a combined thickness of the channel layer **182** and the back barrier layer **180** may have a thickness defined as a distance between an upper surface of the substrate layer **102** and a lower surface of the barrier layer **108**. In one aspect, a combined thickness of the channel layer **182** and the back barrier layer **180** along the Y axis between an upper surface of the channel layer **182** and a lower surface the back barrier layer **180** may be 10%-20%, 20%-30%, 30%-40%, 40%-50%, 50%-60%, 60%-70%, 70%-80%, or 80%-90% of a thickness of the substrate layer **102**. In one aspect, a combined thickness of the channel layer **182** and the back barrier layer **180** may be less than 0.8 microns, less than 0.7 microns, less than 0.6 microns, less than 0.5 microns, or less than 0.4 microns. In one aspect, a combined thickness of the channel layer **182** and the back barrier layer **180** may have a range of 0.8 microns to 0.6 microns, 0.7 microns to 0.5 microns, 0.6 microns to 0.4 microns, 0.5 microns to 0.3 microns, 0.4 microns to 0.2 microns, or 0.7 microns to 0.3 microns. In one aspect, the back barrier layer **180** may be thicker than the channel layer **182** along the Y axis between an upper surface and lower surface of each. In one aspect, the back barrier layer **180** may be 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 120%, 140%, or 160% thicker than the channel layer **182** along the Y axis between an upper surface and lower surface of each. In one aspect, the back barrier layer **180** may be 10%-20%, 20%-30%, 30%-40%, 40%-50%, 50%-60%, 60%-70%, 70%-80%, 80%-90%, 90%-100%, 100%-120%, 120%-140%, or 140%-160% thicker than the channel layer **182** along the Y axis between an upper surface and lower surface of each.

In one aspect, the transistor **100** may have an intervening layer(s) thickness defined as a length between an upper surface of the substrate layer **102** and a lower surface of the barrier layer **108**. In one aspect, the intervening layer(s) thickness may be less than 0.8 microns, less than 0.7 microns, less than 0.6 microns, less than 0.5 microns, or less than 0.4 microns. In one aspect, the intervening layer(s) thickness may have a range of 0.8 microns to 0.6 microns, 0.7 microns to 0.5 microns, 0.6 microns to 0.4 microns, 0.5 microns to 0.3 microns, or 0.4 microns to 0.2 microns.

In one aspect of the transistor **100** as described herein, the p-type material layer **106** may be doped as highly as possible with minimum achievable sheet resistance. In one aspect, the p-type material layer **106** may have an implantation concentration less than 10^{19}. In one aspect, the p-type material layer **106** may have an implantation concentration less than 10^{20}. In one aspect, the p-type material layer **106** may have an implantation concentration of 10^{17}-10^{20}, 10^{19}-10^{20}, 10^{18}-10^{19}, or 10^{17}-10^{18}. In one aspect, the p-type material layer **106** may have an implantation concentration 10^{19} or greater. In one aspect, the p-type material layer **106** may have an implantation concentration of 10^{18}-10^{20}, 10^{18}-10^{19}, or 10^{19}-10^{20}.

In one aspect of the transistor **100** as described herein, the p-type material layer **106** doping may be less than 1E17 cm^3. In one aspect, the p-type material layer **106** doping may be less than 2E17 cm^3. In one aspect, the p-type material layer **106** doping may be less than 6E17 cm^3. In one aspect, the p-type material layer **106** doping may be less than 2E18 cm^3. In one aspect, the p-type material layer **106** doping may be in the range of 5E15 to 5E17 per cm^3. In these aspects, the p-type material layer **106** doping concentration may be greater than a doping concentration of the p-type material layer **106**.

One aspect of the transistor **100** may be implemented as a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages. One aspect includes a method associated with the transistor **100** that may be implemented as a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages. One aspect includes a method of implementing the transistor **100** as a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages. One aspect includes a method of making the transistor **100** as a circuit and Group III-Nitride transistor with buried p-layer and controlled gate voltages.

Accordingly, the disclosure has presented a solution to addressing gate lag effect in Group-III nitride HEMTs and improving the performance of such devices. Additionally, the disclosure has presented a solution to addressing traps that cause memory effects that adversely affect performance. Moreover, the disclosure has set forth a simpler alternative solution to forming p-type layers in HEMTs. The disclosed structure can be readily fabricated with currently available techniques. Moreover, the disclosed use of a high-purity material reduces drain lag effects. Additionally, the disclosed p-type material layer provides a retarding electric field to obtain good electron confinement with low leakage. Additionally, aspects of this disclosure have described in detail variations of transistors with p-type layers and the ways those p-type layers are formed. The disclosed transistors maximize RF power, allow for efficient discharge, and maximize breakdowns.

According to aspects of this disclosure, one or more aspects of the transistor **100** as disclosed may be utilized for high power RF (radio frequency) amplifiers, for high power radiofrequency (RF) applications, and also for low frequency high power switching applications. The advantageous electronic and thermal properties of GaN HEMTs also make them very attractive for switching high power RF signals. In this regard, the disclosure has described a structure with a buried p-layer under the source region to obtain high breakdown voltage in HEMTs for various applications including power amplifiers while at the same time eliminating drifts in device characteristics arising from trapping in the buffer and/or semi-insulating substrates. Use of buried p-layers may also be important in HEMTs for RF switches to obtain high breakdown voltage and good isolation between the input and output.

According to aspects of this disclosure, one or more aspects of the transistor **100** as disclosed may be utilized to implement an amplifier, a radar amplifier, radar components, a microwave radar amplifier, a power module, a gate driver, a component such as a General-Purpose Broadband component, a Telecom component, a L-Band component, a S-Band component, a X-Band component, a C-Band component, a Ku-Band component, a Satellite Communications component, a Doherty configuration, and/or the like. The L band is the Institute of Electrical and Electronics Engineers (IEEE) designation for the range of frequencies in the radio spectrum from 1 to 2 gigahertz (GHz). The S band is a designation by the IEEE for a part of the microwave band of the electromagnetic spectrum covering frequencies from 2 to 4 GHz. The X band is the designation for a band of frequencies in the microwave radio region of the electromagnetic spectrum indefinitely set at approximately 7.0-11.2 GHz. The C-band is the designation given to the radio frequencies from 500 to 1000 MHz. The Ku band is the portion of the electromagnetic spectrum in the microwave range of frequencies from 12 to 18 GHz.

According to aspects of this disclosure, one or more aspects of the transistor **100** as disclosed may be configured in a package and may be implemented as a RF package, a MMIC RF package, and/or the like and may house RF devices. In particular, the RF devices may implement one or more of resistors, inductors, capacitors, Metal-Oxide-Silicon (MOS) capacitors, impedance matching circuits, matching circuits, input matching circuits, output matching circuits, intermediate matching circuits, harmonic filters, harmonic terminations, couplers, baluns, power combiners, power dividers, radio frequency (RF) circuits, radial stub circuits, transmission line circuits, fundamental frequency matching circuits, baseband termination circuits, second order harmonic termination circuits, integrated passive devices (IPD), matching networks, and the like to support various functional technology as input, output, and/or intrastage functions to the package, and/or the like. The package implemented as a MMIC package may further include the transistor **100**. The package implemented as a MMIC package may include, connect, support, or the like a radar transmitter, radar transmitter functions, a microwave radar transmitter, microwave radar transmitter functions, a radar receiver, radar receiver functions, a microwave radar receiver, microwave radar receiver functions, and/or the like.

While the disclosure has been described in terms of exemplary aspects, those skilled in the art will recognize that the disclosure can be practiced with modifications in the spirit and scope of the appended claims. These examples given above are merely illustrative and are not meant to be an exhaustive list of all possible designs, aspects, applications or modifications of the disclosure.

What is claimed is:

1. An apparatus, comprising:
a substrate;
a group III-Nitride barrier layer;
a source electrically coupled to the group III-Nitride barrier layer;
a gate on the group III-Nitride barrier layer;
a drain electrically coupled to the group III-Nitride barrier layer;
an input lead connected to the gate and the input lead configured to receive an input signal;
a p-region being arranged at or below the group III-Nitride barrier layer; and
a gate control circuit configured to control a gate voltage of the gate,
wherein a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain; and
wherein the gate control circuit comprises a voltage source tied to ground.

2. The apparatus of claim **1**
wherein the gate control circuit is configured to reduce a peak negative gate voltage in comparison to operation of the apparatus without the gate control circuit;
wherein the gate control circuit comprises the voltage source tied to ground; and
wherein the apparatus is configured as a radiofrequency package.

3. The apparatus of claim **1**
wherein the gate control circuit is configured to reduce a peak negative gate voltage in comparison to operation of the apparatus without the gate control circuit;
wherein the p-region together with the gate control circuit are configured to at least one of the following: reduce drain lag trapping and reduce gate lag trapping; and
wherein the gate control circuit is tied to ground.

4. The apparatus of claim **1**
wherein the gate control circuit is configured to limit a peak negative gate voltage at the gate in comparison to operation of the apparatus without the gate control circuit.

5. An apparatus, comprising:
a substrate;
a group III-Nitride barrier layer;
a source electrically coupled to the group III-Nitride barrier layer;

a gate on the group III-Nitride barrier layer;
a drain electrically coupled to the group III-Nitride barrier layer;
an input lead connected to the gate and the input lead configured to receive an input signal;
a p-region being arranged at or below the group III-Nitride barrier layer; and
a gate control circuit configured to control a gate voltage of the gate,
wherein a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain;
wherein the gate control circuit is configured to prevent a peak negative gate voltage at the gate from dropping below −8V in greater negative voltage values; and
wherein the apparatus is configured as an amplifier.

6. The apparatus of claim **1**
wherein the gate control circuit is configured to reduce a peak negative gate voltage in comparison to operation of the apparatus without the gate control circuit;
wherein the gate control circuit comprises a diode based circuit configured to compensate for a voltage dependent input capacitance to at least one of the following: reduce drain lag trapping and reduce gate lag trapping; and
wherein the gate control circuit comprises the voltage source tied to ground.

7. The apparatus of claim **1** wherein the gate control circuit comprises a second harmonic short circuit.

8. The apparatus of claim **1** wherein the gate control circuit comprises a second harmonic short circuit tuned and configured to limit a peak negative gate voltage in comparison to operation of the apparatus without the gate control circuit.

9. The apparatus of claim **1**
wherein the gate control circuit is configured to reduce a peak negative gate voltage in comparison to operation of the apparatus without the gate control circuit thereby reducing and/or limiting gate lag trapping; and
wherein the gate control circuit is tied to ground.

10. The apparatus of claim **1**
wherein the p-region together with the gate control circuit are configured to at least one of the following: reduce drain lag and reduce gate lag;
wherein the gate control circuit comprises the voltage source; and
wherein the apparatus is configured as a radiofrequency package.

11. The apparatus of claim **1**
wherein the gate control circuit is configured to shape a waveform of a gate voltage at the gate to inhibit the gate voltage from reaching greater negative voltage values; and
wherein the gate control circuit comprises a diode based circuit configured to compensate for a voltage dependent input capacitance.

12. The apparatus of claim **1**
wherein the gate control circuit is configured to limit a peak negative gate in comparison to operation of the apparatus without the gate control circuit; and
wherein the apparatus is configured as an amplifier.

13. An apparatus, comprising:
a substrate;
a group III-Nitride barrier layer;
a source electrically coupled to the group III-Nitride barrier layer;
a gate on the group III-Nitride barrier layer;
a drain electrically coupled to the group III-Nitride barrier layer;
an input lead connected to the gate and the input lead configured to receive an input signal;
a p-region being arranged at or below the group III-Nitride barrier layer; and
a gate control circuit configured to control a gate voltage of the gate,
wherein a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain;
wherein the gate control circuit is configured with an apparent capacitance as a function of a capacitance of a transistor; and
wherein the apparatus is configured as an amplifier.

14. The apparatus of claim **1**, further comprising:
a group III-Nitride buffer layer on the substrate,
wherein the group III-Nitride barrier layer is arranged on the group III-Nitride buffer layer, the group III-Nitride barrier layer comprising a higher bandgap than a bandgap of the group III-Nitride buffer layer,
wherein the gate control circuit comprises the voltage source tied to ground.

15. The apparatus of claim **1**, further comprising:
a group III-Nitride back barrier layer on the substrate; and
a group III-Nitride channel layer on the group III-Nitride back barrier layer, wherein the group III-Nitride barrier layer is on the group III-Nitride channel layer, the group III-Nitride barrier layer comprising a higher bandgap than a bandgap of the group III-Nitride channel layer.

16. The apparatus of claim **1**, wherein:
a part of a source side of the substrate is free of the p-region;
a part of a drain side of the substrate is free of the p-region;
the source side extends from the gate toward and past the source;
the drain side extends from the gate toward and past the drain; and
the gate control circuit comprises the voltage source tied to ground.

17. An apparatus, comprising:
a substrate;
a group III-Nitride barrier layer;
a source electrically coupled to the group III-Nitride barrier layer;
a gate on the group III-Nitride barrier layer;
a drain electrically coupled to the group III-Nitride barrier layer;
an input lead connected to the gate and the input lead configured to receive an input signal;
a p-region being arranged at or below the group III-Nitride barrier layer;
a gate control circuit configured to control a gate voltage of the gate; and
a field plate,
wherein a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain;
wherein the p-region is implanted; and
wherein the apparatus is configured as an amplifier.

18. An apparatus, comprising:
a substrate;
a group III-Nitride barrier layer;
a source electrically coupled to the group III-Nitride barrier layer;

a gate on the group III-Nitride barrier layer;

a drain electrically coupled to the group III-Nitride barrier layer;

an input lead connected to the gate and the input lead configured to receive an input signal;

a p-region being arranged at or below the group III-Nitride barrier layer;

a gate control circuit configured to control a gate voltage of the gate; and

a field plate,

wherein a portion of the p-region is arranged vertically below at least one of the following: the source, the gate, and an area between the gate and the drain;

wherein the field plate is electrically coupled to the source; and

wherein the apparatus is configured as an amplifier.

19. The apparatus of claim **1** wherein:

a distance LGPS defines a length of a portion of the p-region from a lower corner of the gate on a source side toward the source;

a distance LGPD defines a length of a portion of the p-region from a lower corner of the gate on a drain side toward the drain; and

the distance LGPS greater than the distance LGPD,

wherein the source side extends from the gate toward and past the source; and

wherein the drain side extends from the gate toward and past the drain.

20. The apparatus of claim **1** wherein the gate control circuit comprises a second harmonic short circuit tuned and configured to reduce and/or limit peak gate voltages.

21. The apparatus of claim **1** wherein the gate control circuit comprises a resonant circuit.

22. The apparatus of claim **1** wherein the gate control circuit comprises a second harmonic short circuit tied to ground.

23. The apparatus of claim **1**

wherein the gate control circuit comprises a diode-based circuit; and

wherein the apparatus is configured as a radiofrequency package.

24. The apparatus of claim **1**

wherein the gate control circuit comprises a second harmonic short circuit tied to ground; and

wherein the gate control circuit comprises a diode-based circuit comprising a diode configured to counteract non-linear input capacitance.

25. The apparatus of claim **1** wherein the gate control circuit comprises a second harmonic short circuit tied to ground;

wherein the gate control circuit comprises a diode-based circuit comprising a diode configured to counteract non-linear input capacitance; and

wherein the gate control circuit comprises the voltage source tied to ground.

26. The apparatus of claim **1** comprising a transistor package and the apparatus further comprising:

an input matching network connected between the input lead and the gate; and

the gate control circuit is connected between the input matching network and the gate,

wherein the gate control circuit comprises the voltage source tied to ground.

27. The apparatus of claim **1** further comprising an input matching network, wherein the gate control circuit is implemented as part of the input matching network,

wherein the gate control circuit is tied to ground.

28. The apparatus of claim **1** comprising a transistor package and the apparatus further comprising:

an input matching network connected between the input lead and the gate;

the gate control circuit is connected between the input matching network and the gate; and

an output matching network connected between an output lead and the source,

wherein the gate control circuit comprises the voltage source tied to ground.

29. The apparatus of claim **28**,

wherein the output matching network comprises one or more inductances, impedances, and/or capacitors connected between an output lead and the source; and

wherein the apparatus is configured as an amplifier.

30. The apparatus of claim **28** further comprising an output matching network, wherein the output matching network comprises one or more inductances, impedances, and/or capacitors connected between an output lead and the source,

wherein the apparatus is configured as an amplifier.

31. The apparatus of claim **28**

wherein the input matching network comprises one or more inductances, impedances, and/or capacitors connected between the input lead and the gate; and

wherein the apparatus is configured as an amplifier.

32. The apparatus of claim **28** wherein the gate control circuit comprises a second harmonic short circuit tuned and configured to reduce and/or limit peak gate voltages.

33. The apparatus of claim **28** wherein the gate control circuit comprises a resonant circuit.

34. The apparatus of claim **28** wherein the gate control circuit comprises a second harmonic short circuit tied to ground.

35. The apparatus of claim **28**

wherein the gate control circuit comprises a diode-based circuit; and

wherein the apparatus is configured as an amplifier.

36. The apparatus of claim **28**

wherein the gate control circuit comprises the voltage source tied to ground; and

wherein the apparatus is configured as an amplifier.

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